A MODULAR AND COST-EFFECTIVE FRAMEWORK FOR WIRELESS FOUNDATION MODELS THROUGH WORLD KNOWLEDGE INTEGRATION

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Abstract

THE development of foundation models for wireless communications requires two key components: a powerful signal encoder to extract rich features and an intelligent reasoning model for coherent, cross-task decision-making. As building such models from scratch is prohibitively expensive, this dissertation presents a practical framework that creates robust wireless foundation models by strategically approximating and integrating world knowledge.

Our cost-effective approach proposes a modular, three-component framework and provides proof-of-concept validation for each component. The first is a rich feature encoder, for which we develop DenoMAE. Through unsupervised learning on unlabeled signals, these models approximate world knowledge of universal signal representations by treating noise as an informative modality and processing multimodal time-domain and constellation-diagram data. The second is an intelligent reasoning engine, for which we demonstrate a novel proof-of-concept by utilizing LLMs to perform open-set classification based on statistical signal features, successfully transferring their pre-trained world knowledge to the wireless domain. The third is an adaptation module, for which we employ Meta-Task learning to enable the framework to rapidly apply its integrated world knowledge to new tasks and channel conditions with minimal labeled data.

This dissertation demonstrates that the key to economical and effective wireless foundation models lies not in massive, isolated training, but in the design of this strategic, modular

framework. By validating the components for unsupervised representation (DenoMAE), LLM-based reasoning, and meta-learning for adaptation, our work provides a practical pathway and a conceptual blueprint for achieving foundation model capabilities in wireless communications.

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Nomenclature

AI Artificial Intelligence

ANN Artificial Neural Network

API Application Programming Interface

DenoMAE Denoising Masked Autoencoder

DL Deep Learning

GAN Generative Adversarial Network

GPU Graphics Processing Unit

LLM Large Language Model

MAE Masked Autoencoder

ML Machine Learning

MSE Mean Squared Error

NLP Natural Language Processing

ReLU Rectified Linear Unit

RL Reinforcement Learning

SGD Stochastic Gradient Descent

t-SNE t-Distributed Stochastic Neighbor Embedding

UML Unified Modeling Language

WPFM Wireless Physical-Layer Foundation Model

Chapter 1

Introduction

THE rapid evolution of wireless communications, driven by technologies such as the upcoming 6G standard[2], necessitates a transformation in how Artificial Intelligence (AI) addresses challenges unattainable by traditional optimization techniques. The remarkable success of foundation models in computer vision (CLIP[3], MAE[4]) and natural language processing (GPT[5], BERT[6]) has demonstrated the transformative power of learning universal representations that generalize across tasks with minimal fine-tuning. These models leverage massive unlabeled datasets through self-supervised learning to discover rich, transferable features that encode fundamental properties of their respective domains. Yet, the current AI landscape in wireless networks remains dominated by models developed and trained for isolated, task-specific applications[2], leading to significant limitations in knowledge sharing, reuse, and adaptability. This dissertation addresses a fundamental question: *Can we develop practical pathways toward Wireless Physical-Layer Foundation Models (WPFMs) by strategically approximating and integrating world knowledge through modular, cost-effective architectures?*

The limitations of current wireless AI systems are particularly evident in Automatic Modulation Classification (AMC)—a critical component of spectrum management, signal

identification, and cognitive radio systems[7]. Conventional deep learning approaches face three major challenges: (1) **Imbalance across conditions:** Models often fail to distinguish between low and high Signal-to-Noise Ratio (SNR) conditions[7], resulting in uneven classification performance where low SNR signals obscured by noise require specialized handling compared to high SNR signals; (2) **Data scarcity and training overhead:** Acquiring high-quality labeled radio frequency (RF) signal data demands specialized expertise and equipment, while traditional deep learning necessitates large labeled datasets[8], creating barriers to scalability; (3) **Lack of interpretability and generalization:** Task-specific models struggle to adapt quickly to new tasks, generalize across diverse channel characteristics, and lack capacity for human-understandable interaction[7].

Building wireless foundation models from scratch to address these challenges is prohibitively expensive [3]. Unlike images or text, wireless signals exhibit unique characteristics: they are inherently noisy, their structure depends on both transmitter design and channel physics, and their semantic meaning is encoded in subtle statistical patterns. These properties demand an economical approach that leverages existing world knowledge rather than massive isolated training. This dissertation presents such a solution: a modular and cost-effective framework that achieves foundation model capabilities through the strategic integration of world knowledge. Our approach validates three essential components that work synergistically: (1) a **rich feature encoder** (DenoMAE) that uses self-supervised learning to approximate world knowledge of signal representations; (2) an **intelligent reasoning engine** (LLM integration) that performs open-set classification by converting signal features into natural language prompts; and (3) an **adaptation module** (Meta-Task) that enables rapid knowledge transfer to new tasks with minimal data. By demonstrating proof-of-concept for these core components, we provide both a practical framework and a conceptual blueprint for achieving foundation model capabilities in wireless communications without prohibitive training costs.

1.1 Objectives and Motivation

The central objective of this dissertation is to establish foundational principles and practical methodologies for developing Wireless Physical-Layer Foundation Models (WPFMs)—universal representations of wireless signals that enable generalization across modulation schemes, channel conditions, and tasks. Rather than pursuing the cost-prohibitive strategy of building massive models from scratch, this work demonstrates that foundation model capabilities can be achieved economically through a strategic, modular design that combines self-supervised learning, LLM-based reasoning, and meta-learning adaptation.

The motivation for this approach stems from the success of foundation models in other domains. Models like GPT-4 [9] and CLIP [3] have shown that pre-training on broad data distributions develops powerful, transferable representations. These successes rely on three key ingredients: massive unlabeled datasets, self-supervised pretext tasks that capture domain-invariant structure, and flexible architectures that enable task-specific adaptation. While wireless communications generates petabytes of unlabeled signal data daily, its unique characteristics—the noisy, physics-dependent nature of signals—present distinct challenges. This leads to the central question driving this dissertation: *How can we build WPFMs that understand and represent these complex signals without the prohibitive costs of isolated training, while still achieving the generalization capabilities of their vision and language counterparts?*

To address this, the remainder of this section is structured to first articulate the long-term vision for WPFMs. It then deconstructs this vision into a set of five core research questions that guide the technical work. Finally, it outlines the specific technical approach and contributions developed in this dissertation to answer these questions.

1.1.1 Vision: Wireless Foundation Models

We envision wireless foundation models analogous to CLIP in vision or GPT in language: pre-trained on diverse, unlabeled signals to learn representations encoding fundamental wireless signal properties, then adapted to specific tasks (classification, denoising, channel estimation) with minimal fine-tuning. Such models would:

- **Generalize Across Modulations:** Recognize modulation schemes never encountered during training through compositional understanding of signal properties, enabled by rich learned representations
- **Handle SNR Imbalance Robustly:** Employ hybrid generative-discriminative learning (DenoMAE) combined with robust discriminative classifiers (improved plug-and-play), ensuring consistent performance from -10 dB to 30+ dB
- **Operate Cost-Effectively:** Enable training-free or few-shot deployment through LLM integration and self-supervised pre-training, avoiding exhaustive retraining for each scenario
- **Adapt Rapidly with Minimal Data:** Leverage meta-learning to quickly specialize for new deployment scenarios with few labeled examples, dramatically reducing data requirements
- **Provide Interpretable Reasoning:** Convert signal features to natural language prompts for LLM-based classification, enabling human-understandable explanations and prompt-based optimization
- **Leverage World Knowledge:** Integrate pre-trained LLMs' reasoning capabilities to approximate and transfer existing knowledge rather than learning everything from scratch

1.1.2 Research Questions

This dissertation addresses three interconnected research questions toward realizing cost-effective wireless foundation models:

RQ1: Can vision-based, self-supervised learning effectively transfer world knowledge to the wireless domain?

- How can wireless signals be represented as images (e.g., spectrograms, constellation diagrams) to leverage pre-trained vision models like ViT and YOLO?
- What self-supervised objectives (e.g., masked autoencoding) are most effective for learning rich, transferable representations from these signal-images?
- Can multimodal learning, combining different image representations (e.g., IQ samples and constellation diagrams), improve feature quality and robustness?
- How effectively do features learned by vision transformers transfer to wireless-specific tasks like modulation classification and signal detection?
- To what extent can this cross-modal approach reduce the dependency on large labeled datasets in the wireless domain?

RQ2: Can LLMs enable open-set classification of wireless signals through world knowledge transfer?

- Can statistical signal features (cumulants, spectral properties) be converted to natural language prompts for LLM reasoning?
- Do pre-trained LLMs' in-context learning capabilities transfer to wireless signal classification without fine-tuning?

- What bridges the semantic gap between quantitative signal properties and linguistic descriptions?
- How competitive is training-free LLM-based classification compared to extensively trained task-specific models?
- Can LLMs provide interpretable reasoning for wireless classification decisions, enabling human-understandable explanations?

RQ3: Can meta-learning enable rapid adaptation to new tasks with minimal labeled data?

- What episodic training strategies maximize generalization for wireless classification tasks?
- Can task-aware regularization within meta-learning prevent overfitting while maintaining adaptability?
- How can we design episode generation strategies that account for signal-specific characteristics like SNR variation and modulation similarity?
- What design principles enable efficient knowledge transfer across different channel conditions and modulation schemes?
- How many labeled examples are minimally required for effective adaptation to new wireless tasks?

1.2 Scope

This work focuses on automatic modulation classification (AMC) as the primary testbed for developing foundational principles for wireless physical-layer foundation models. The

scope is defined by the following domain parameters and limitations.

1.2.1 Primary Domain: Wireless Communication

Our core application domain is automatic modulation classification under challenging conditions:

- **Signal Representations:** We explore time-domain IQ samples, constellation diagrams, and noise patterns as complementary modalities
- **Operating Conditions:** SNR ranges from -10 dB to 30 dB, covering severe noise to clean signal regimes[]
- **Modulation Schemes:** Focus on common digital modulations (BPSK, QPSK, 8PSK, QAM variants, FSK, GFSK)[] with extensions to emerging schemes
- **Channel Models:** Primarily AWGN and fading channels[] representative of practical wireless systems
- **Datasets:** synthetic datasets for controlled experiments

1.2.2 Scope Limitations

While this work establishes foundations for wireless AI, certain aspects remain future work:

- **End-to-End Communication:** Focus is on physical layer modulation classification, not complete communication system design[]
- **Hardware Implementation:** Validation uses simulated and standard datasets rather than custom hardware deployments[]

- **Adversarial Robustness:** While noise robustness is central, adversarial attacks[] and countermeasures are not extensively explored
- **Multi-User/MIMO Scenarios:** Primarily single-user, single-antenna scenarios; multi-antenna[] and interference management are secondary
- **Foundation Model Scale:** Due to computational constraints, models are smaller than vision/language foundation models[]; principles established here enable future scaling

1.3 Contributions

This dissertation makes four primary technical contributions that collectively establish a modular, cost-effective pathway toward Wireless Physical-Layer Foundation Models (WPFMs). The first three contributions form the core components of the proposed framework, while the fourth provides a crucial benchmark and methodology for cross-modal knowledge transfer.

1.3.1 Toward Wireless Foundation Models: A Modular, Cost-Effective Framework

Central Thesis Contribution: We propose a modular and cost-effective framework for Wireless Physical-Layer Foundation Models (WPFMs) that strategically approximates and integrates world knowledge, aiming to demonstrate that:

- **Self-Supervised Learning Component:** Multimodal masked autoencoders (DenoMAE) will learn universal wireless signal representations through unsupervised learning on unlabeled data, dramatically reducing labeled data requirements (80% reduction) while approximating world knowledge of signal patterns

- **LLM Integration Component:** Large language models can perform training-free, one-shot wireless classification through a novel proof-of-concept that converts statistical signal features to natural language prompts, transferring pre-trained language model reasoning to wireless domain without wireless-specific training

- **Hybrid Architecture Component:** DenoMAE 's hybrid generative-discriminative learning combined with an improved plug-and-play discriminative classifier will ensure robust performance across SNR conditions (-10 dB to 30+ dB), addressing the critical SNR imbalance problem through rich representation learning and condition-aware classification

- **Meta-Learning Component:** Meta-Task learning will enable rapid adaptation to novel modulation types with minimal labeled data (5-10 examples), efficiently applying integrated knowledge to new tasks and channel conditions

- **Cost-Effectiveness Principle:** The key to economical wireless foundation models lies in this strategic, modular framework design that leverages existing world knowledge rather than massive isolated training

This modular approach will provide both a conceptual blueprint and practical pathway toward next-generation wireless AI systems that are scalable, interpretable, and economically deployable. The framework's modularity will enable: (1) independent component optimization and updates; (2) targeted improvements without full system retraining; (3) flexible integration of future advances in self-supervised learning, LLMs, or meta-learning; (4)

cost-effective deployment through training-free operation and minimal data requirements.

1.4 Structure

This dissertation is organized to guide readers from foundational concepts through technical contributions toward the unifying vision of wireless foundation models. The thesis follows a standard academic structure with nine chapters, each building progressively on previous content.

- **Chapter 2 and 3: Self-Supervised Representation Learning (DenoMAE & CDRF):** Presents Contribution 1, addressing RQ1, RQ4, and RQ5. Introduces DenoMAE, a multimodal masked autoencoder that treats noise as an informative modality. Details the architecture processing time-domain IQ samples and constellation diagrams through vision transformers. This chapter also introduces the CDRF benchmark, demonstrating the successful application of computer vision models to RF drone detection by representing signals as images. This combined work establishes that vision transformers can approximate world knowledge for wireless signals, enabling significant data reduction and cross-modal understanding.

- **Chapter 4: LLM-Based Training-Free Classification (Plug-and-Play):** Presents Contribution 2, addressing RQ2. Introduces the Plug-and-Play framework that enables training-free wireless classification through LLM reasoning. Details the statistical feature-to-language prompt conversion process. Demonstrates competitive performance with pre-trained LLMs without wireless-specific training, validating cross-domain knowledge transfer from language models.

- **Chapter 5: Meta-Learning for Rapid Adaptation (Meta-Task):** Presents Contribution 3, addressing RQ3. Introduces Meta-Task, which reformulates regularization

as a learnable task within few-shot learning. Details the Task-Decoder architecture and demonstrates significant performance gains and faster convergence, establishing that meta-learning enables rapid adaptation to new wireless tasks with minimal labeled data.

- **Chapter 6: Experimental Results and Analysis:** Consolidates and compares results across all three contributions. Presents comprehensive experimental validation including ablation studies, cross-dataset evaluation, and comparative analysis. Analyzes how each method performs across different SNR conditions, dataset sizes, and task complexities. Discusses performance trade-offs and computational efficiency. Evaluates the potential for component integration and synergies within the modular framework.

- **Chapter 7: Conclusions and Future Work:** Synthesizes key findings and answers the three research questions. Summarizes how DenoMAE, Plug-and-Play, and Meta-Task provide complementary pathways toward wireless foundation models. Discusses remaining challenges and outlines concrete future research directions, including scaling to larger models, extending to multi-antenna scenarios, and exploring generative capabilities. Provides preliminary discussion of ongoing work (DenoMAE 2.0, DISC-AMC) as natural extensions. Concludes with broader impact considerations for next-generation wireless AI systems.

Chapter 2

DenoMAE: A Multimodal Autoencoder for Denoising Modulation Signals

2.1 Introduction

In recent years, the deep learning paradigm has shifted to address the high costs and scarcity of labeled data and the intensive data demands of supervised models. Self-supervised learning has emerged as an efficient solution, allowing models to leverage abundant raw data in pretraining. This approach drastically reduces the need for labeled data during fine-tuning while maintaining strong downstream performance [10]. Self-supervised methods have shown remarkable success across fields like natural language processing [11] and computer vision [12, 3], paving the way for new applications in fields with sparsely labeled data, including communications.

In communication systems, particularly wireless networks, maintaining signal integrity is crucial, yet it's often compromised by limited data availability and varying noise levels during transmission. Estimating modulation signals becomes especially challenging

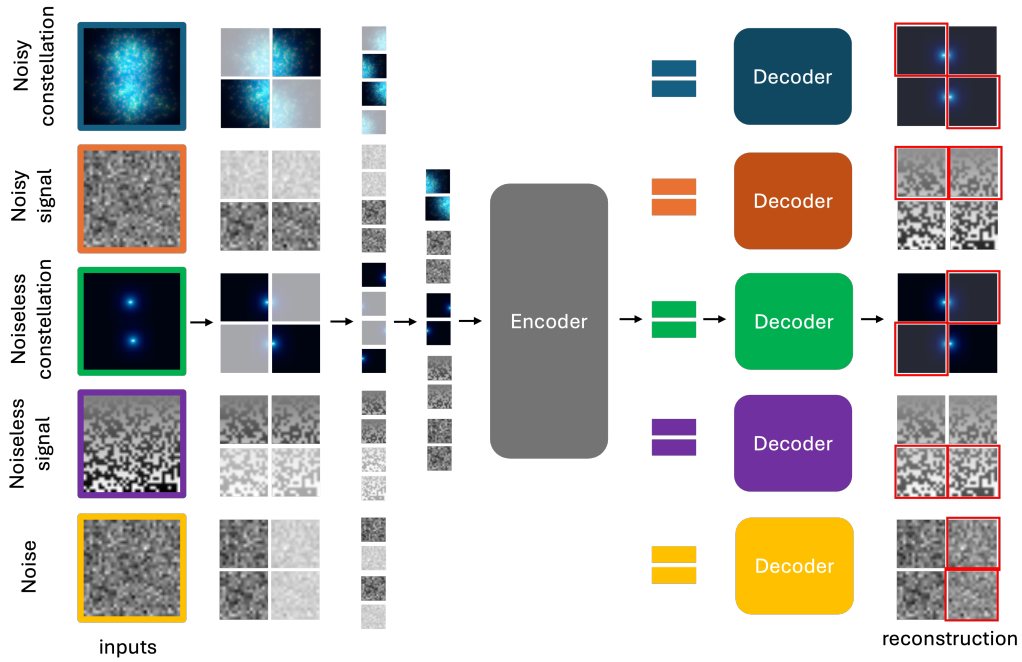


Figure 2.1: **DenoMAE Pretraining Strategy**: We apply a random 75% masking (not to scale in the illustration) across all input modalities. The remaining 25% of visible patches are processed by a shared encoder, while each modality utilizes a dedicated decoder to reconstruct its masked patches. Only the encoder is reused for fine-tuning the downstream tasks.

as noise levels fluctuate with changing SNRs [13]. This limitation becomes particularly problematic when models encounter previously unseen modulation types [14]. While existing models can often classify known modulation schemes, they struggle to generalize to new or noisy conditions, resulting in misclassification and necessitating extensive retraining [15, 16]. This problem highlights an urgent need for robust, adaptive models that can generalize across diverse conditions with minimal reliance on labeled data.

One approach to improving generalizability and robustness is through pretraining methods that not only capture robust representations but also denoise signals effectively, enhancing classification accuracy in downstream tasks and providing scalable solutions in dynamic communication environments. Masking-based methods have traditionally been

effective for denoising and improving signal clarity, but they are often limited to single-modality approaches. Though effective, these methods are inherently constrained by the information within their specific modality, which restricts their potential to leverage cross-modal interactions for more robust learning [17, 18]. Recent advances in multimodal learning address this limitation by integrating data from multiple modalities, enhancing model robustness and enabling more flexible zero-shot learning capabilities. In works like [19, 20], attention mechanisms and masked modeling pretraining across modalities have successfully reconstructed masked signals, proving beneficial in improving classification under complex conditions [21, 22].

Building on recent advancements, we adapt these techniques and present DenoMAE, a multimodal autoencoder tailored for the unique challenges in communication systems. DenoMAE's innovative approach involves treating noise as an additional modality, allowing the model to generate all other modalities from a single input modality. This unique framework helps the network gain a comprehensive understanding of noise across modalities, significantly enhancing its denoising capabilities. A random masking strategy, which hides 75% of the input modalities, is employed to challenge the model in reconstructing the missing patches, reducing computational complexity, and introducing beneficial uncertainty conducive to effective learning.

Within the broader framework of this dissertation, DenoMAE serves as the foundational representation learning component. While Chapter 4 addresses training-free classification through LLMs and Chapter 5 enables rapid adaptation, DenoMAE provides the robust feature encoder that learns universal signal representations from minimal labeled data. This modular approach demonstrates that wireless foundation models need not follow the massive-scale paradigm of vision and language models, but can achieve strong performance through efficient self-supervised pretraining.

Our contributions to DenoMAE are threefold:

- We propose a novel multimodal paradigm that incorporates noise as an additional modality, enabling more effective pretraining on modulation signals and enhancing model resilience to noise.

- We demonstrate that our pre-trained model achieves significant improvements in classification accuracy under low SNR conditions, with accuracy gains of up to 22.1% over non-pre-trained models.

- Our approach outperforms traditional methods, achieving superior results while requiring only one-tenth of the unlabeled data for pretraining and one-third of the labeled data for fine-tuning, underscoring its efficiency and scalability for real-world applications.

Framework Position and Research Questions: Within the dissertation's modular framework for wireless foundation models, DenoMAE serves as the critical representation learning component, directly addressing **RQ1**: *Can vision-based, self-supervised learning effectively transfer world knowledge to the wireless domain?* By treating wireless signals as images (constellation diagrams) and applying masked autoencoding—a technique proven successful in computer vision [23]—DenoMAE demonstrates successful cross-modal knowledge transfer. The novel inclusion of noise as an explicit modality enables the encoder to learn robust, denoised representations from minimal labeled data (80% reduction), validating that vision transformers can approximate world knowledge for wireless signals. This efficient encoder provides the feature extraction backbone that complements the LLM-based reasoning engine (Chapter 4) and enables the rapid adaptation mechanisms of Meta-Task (Chapter 5). Together with CDRF's cross-modal validation (Chapter 3), DenoMAE establishes that cost-effective wireless foundation models can be built through strategic modular design rather than massive isolated training.

2.2 Related Work

For a comprehensive review of modulation classification and multisensory masked modeling approaches, please refer to Chapter **??**, Section 3.2. Here we briefly position our work within this landscape.

Our approach builds upon recent advances in self-supervised learning, particularly masked autoencoders (MAE) [23] and multimodal learning [19]. While existing methods in AMC rely on CNNs [24], transformers [8], or single-modality approaches, DenoMAE introduces a novel paradigm that treats noise as an additional modality, enabling more effective pretraining and enhanced robustness across diverse SNR conditions.

2.3 Methodology

In this work, we propose DenoMAE, an innovative pretraining method for modulation classification that leverages recent advancements in multisensory pretraining. By incorporating noise as an additional modality during pretraining, DenoMAE significantly reduces data requirements while enhancing performance in downstream classification tasks.

2.3.1 DenoMAE Framework

DenoMAE consists of four primary components: input embedding, encoder, shared latent space, and decoder. Figure 2.1 illustrates the overall architecture.

2.3.1.1 Input Embedding

For each modality $m \in 1, ..., n$, we represent the input as a C-channel image $X_m \in \mathbb{R}^{C \times H \times W}$. We divide each image into non-overlapping patches of size $p \times p$, resulting in $N = (H/p)^2$ patches per modality. Each patch is flattened and linearly projected to a

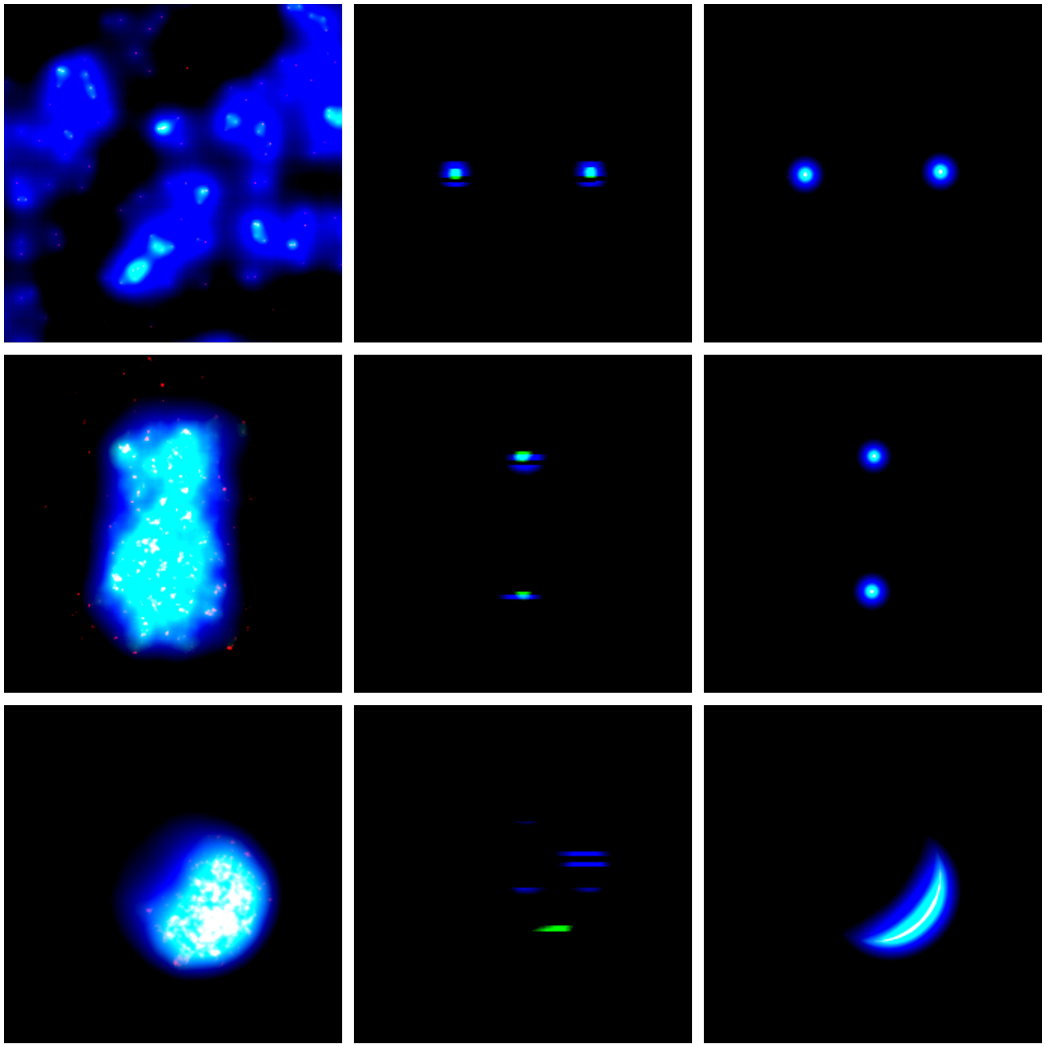


Figure 2.2: Denoised outputs of DenoMAE on unlabeled constellation diagrams at different SNRs during pretraining.

D-dimensional embedding space:

$$E_m = [e_1^m, ..., e_N^m] + E_{pos}, \quad e_i^m \in \mathbb{R}^D$$

where E_{pos} represents learnable positional embeddings.

2.3.1.2 Encoder

The encoder f_{enc} is a series of L transformer blocks. We randomly mask a high proportion (e.g., 75%) of the input patches for each modality. The encoder processes the visible patches:

$$H_m = f_{enc}([e^m_{i_1}, ..., e^m_{i_k}]), \quad i_j \in \text{visible patches}$$

2.3.1.3 Shared Latent Space

To enable cross-modal learning, we project the encoded representations from all modalities into a shared latent space. We use a linear projection followed by layer normalization:

$$Z_m = \text{LN}(W_m H_m + b_m)$$

The shared latent representation Z is the concatenation of all modality-specific projections:

$$Z = [Z_1; ...; Z_5]$$

2.3.1.4 Decoder

The decoder f_{dec} reconstructs the masked patches for all modalities simultaneously. It takes the shared latent representation Z as input and processes it through a series of transformer blocks. The output is then split into modality-specific representations and projected back to the original patch space:

$$\hat{X}m = fproj_m(f_{dec}(Z))$$

where f_{proj_m} is a modality-specific projection layer.

2.3.1.5 Loss Function

We employ a multi-modal reconstruction loss that combines individual losses for each modality:

$$\mathcal{L} = \sum_{m=1}^{5} w_m \mathcal{L}_m$$

where w_m is the weight for modality m, and $\mathcal{L}_m$ is the reconstruction loss for that modality. We use the mean squared error (MSE) as the reconstruction loss:

$$\mathcal{L}m = \frac{1}{|\mathcal{M}m|} \sum i \in \mathcal{M}m|\hat{X}m, i - Xm, i|_2^2$$

2.4 Experimental Setup

We begin this section by detailing the parameters used for sample generation, followed by the classifier's configuration. Finally, we present the parameters applied to our downstream tasks.

2.4.1 Constellation Diagram Generation

Constellation diagrams offer a more informative approach to modulation classification compared to raw time-series signals, capturing richer detail essential for accurate interpretation [25, 26]. The process of generating constellation diagrams for modulation signal classification involves multiple stages. We begin by mapping modulated signals onto a 7×7 complex plane, which provides sufficient space to capture signal samples while maintaining computational efficiency for SNR ranges of -10 dB to 10 dB. The basic constellation diagram is then enhanced through a multi-step process: first, as a gray image that handles varying pixel densities where multiple samples may occupy single pixels; second, as an enhanced grayscale image that employs an exponential decay model ($B_{i,j}$) to account for

both the precise position of samples within pixels and their influence on neighboring pixels. This model considers the sample point's power (P), the distance between sample points and pixel centroids ($d_{i,j}$), and an exponential decay rate (α) [27]. To make the representation compatible with DenoMAE, which expects RGB input, we generate a three-channel image by creating three distinct enhanced grayscale images from the same data samples, each utilizing different exponential decay rates.

2.4.2 Parameters in Sample Generation

The dataset comprises signals from ten modulation formats, each originally of length $L_0 = 1024$. To conform to the transformer's input dimensions, signals undergo a two-step pre-processing: (1) reshaping to $\mathbf{S}_1 \in \mathbb{R}^{32\times32}$, and (2) interpolation to $\mathbf{S}2 \in \mathbb{R}^{224\times224}$. For formats with $L_0 > 1024$ (e.g., $L_{\text{GMSK}} = 8196$), signals are initially downsampled to L_0. To match the three-channel structure of constellation images $\mathbf{I} \in \mathbb{R}^{3\times224\times224}$, $\mathbf{S}_2$ is replicated across three channels, yielding $\mathbf{S}_3 \in \mathbb{R}^{3\times224\times224}$. All signals are sampled at $f_s = 200$ kHz, ensuring consistency across modulation formats.

2.4.3 Parameters for DenoMAE

The MultiMAE architecture processes n =5 modalities of input images $\mathbf{X} \in \mathbb{R}^{3\times224\times224}$, segmented into 16×16 patches. The model employs a transformer-based encoder-decoder structure with $d_{\text{model}} = 768$ embedding dimensions. The encoder consists of $L_e = 12$ layers, while the decoder uses $L_d = 4$ layers, both utilizing $h = 12$ attention heads. During training, a random masking strategy with $p_{\text{mask}} = 0.75$ is applied. The model is optimized using AdamW with learning rate $\eta = 10^{-4}$ over $E = 100$ epochs, using batches of size $B = 64$. This flexible architecture allows for easy adaptation to various multi-modal denoising tasks.

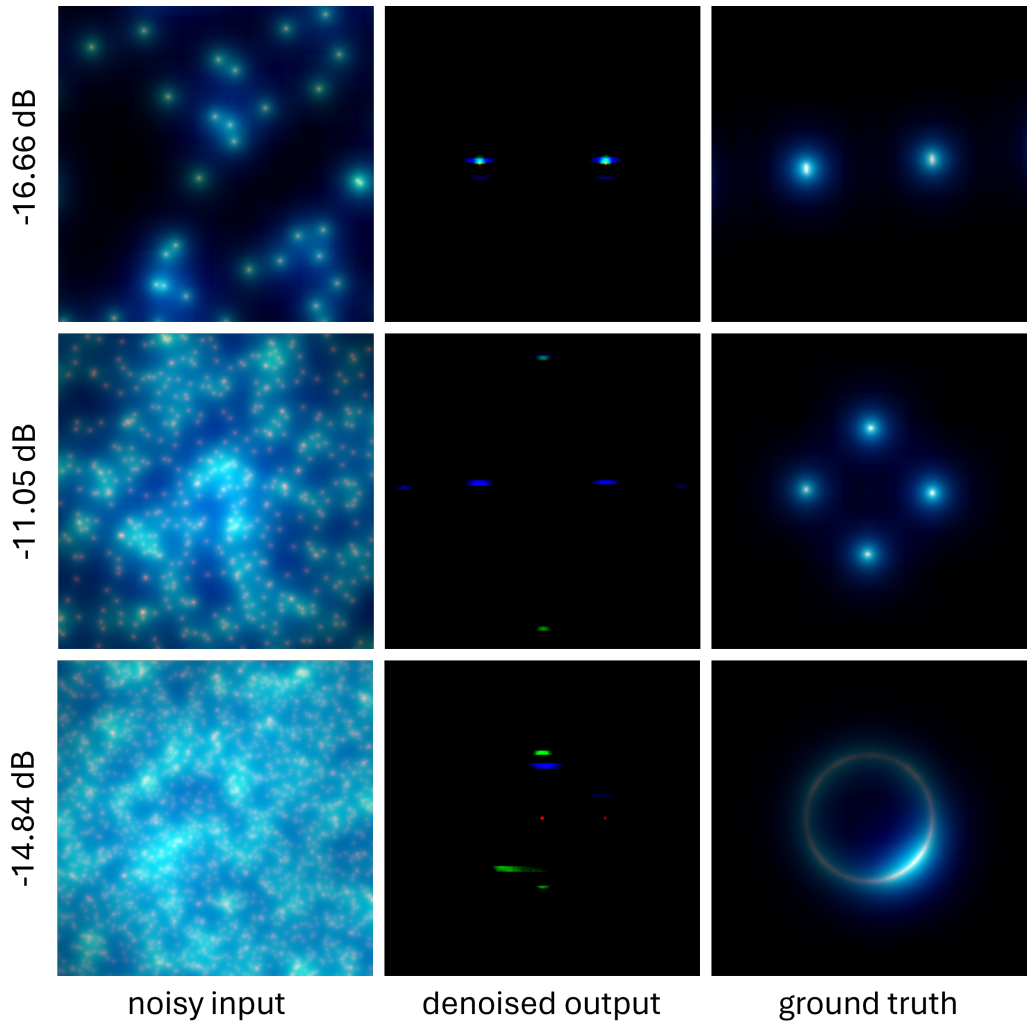


Figure 2.3: Extrapolation-ability of DenoMAE on out-of-bound much lower SNRs.

2.4.4 Finetuning Parameters

The fine-tuning task adapts the pre-trained DenoMAE model for classification across $C = 10$ classes. The encoder $E(\cdot)$ from DenoMAE, with $d_{\text{model}} = 768$, is used as a backbone, followed by a classifier head $H(\cdot)$ consisting of a two-layer MLP with hidden dimension 512 and dropout $p = 0.5$. For an input $\mathbf{X} \in \mathbb{R}^{3\times224\times224}$, the model computes $\hat{y} = H(G(E(\mathbf{X})))$, where $G(\cdot)$ is global average pooling. The model is trained for $E = 150$

Table 2.1: Performance Comparison Across Modalities in Downstream Task

Constellation		Signal		Noise	Test Accuracy (%)
Clean	Noisy	Clean	Noisy		
✓					81.30
✓	✓				81.90
✓	✓	✓			83.20
✓	✓	✓	✓		83.30
✓	✓	✓	✓	✓	83.50

epochs using Adam optimizer with learning rate $\eta = 10^{-4}$ and batch size $B = 32$. Training data undergoes normalization but no augmentation.

2.5 Results and Discussion

This work centers on denoising and reconstructing constellation diagrams, a modality used exclusively during downstream classification tasks. In the following sections, we provide detailed insights into those findings.

Table 2.2: Comparison of DenoMAE with other modulation classification methods

Methods	**Number of samples**		**SNR dB**	**Number of classes**	**Test accuracy (%)**
	Pretraining	**Fine-tuning**			
AlexNet [27]	800,000 (train)	8,000 (test)	2	8	82.8
NMformer	106,800	3,000	0.5 - 4.5	10	71.6
CNN-AMC [28]	79,200	228,060	-6	4	50.40
DL-GRF [29]	2,000 (train)	200 (test)	0	4	59
DenoMAE (**Ours**)	10,000	1,000	0.5	10	83.5
DenoMAE (**Ours**)	10,000	1,000	-10	10	77.50

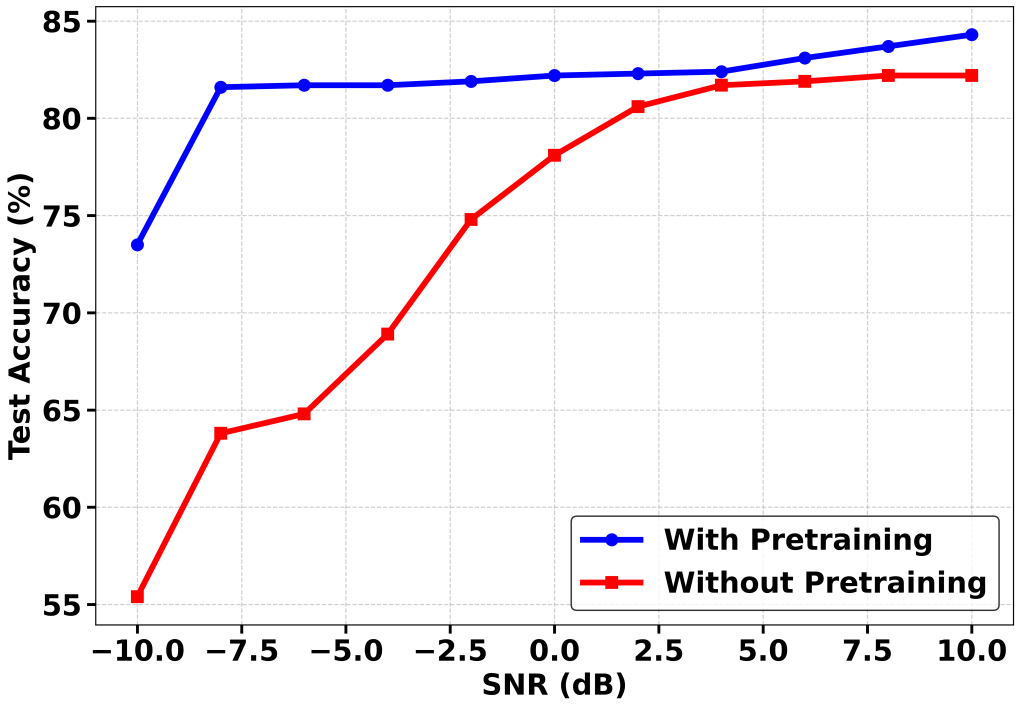


Figure 2.4: DenoMAE fine-tuned classification accuracy for constellation diagrams at different SNRs.

2.5.1 Pretraining Performance

The pretraining effectiveness of DenoMAE was evaluated using 10,000 unlabeled signal samples across five modalities. Visual inspection of the denoised constellation diagram outputs (Figure 2.2) demonstrates strong reconstruction capabilities, particularly for samples in the first two rows where the model successfully removes noise while preserving signal characteristics. While reconstruction quality slightly diminishes for samples in the last row, the overall shape and pattern retention remain sufficient for downstream tasks.

2.5.2 Impact of SNR

In fine-tuning, we used 1000 labeled constellation images with 10 classes. The impact of SNR on classification performance reveals DenoMAE's robust noise resilience. As shown in Figure 2.4, the pre-trained model achieves 77.50% accuracy even at -10 dB SNR, with performance steadily improving to 84.30% at 10 dB. The pretraining advantage is most pronounced in challenging low-SNR conditions, demonstrating a 22.1% improvement over the non-pre-trained baseline at -10 dB. This performance gap gradually narrows at higher SNRs, suggesting that pretraining is particularly valuable for learning noise-robust features.

2.5.3 Extrapolation on out-of-bound SNRs

DenoMAE demonstrates impressive extrapolation capabilities when evaluated on SNR conditions significantly outside its training range. Although trained only on SNR levels above -10 dB, the model continues to deliver meaningful reconstruction performance at challenging SNR levels between -11 dB and -20 dB. As shown in Figure 2.3, DenoMAE effectively denoises highly noisy input constellation diagrams, making the first two samples visually recognizable. The final sample, while impacted by extreme noise, still approximates the outer shape of the ground truth.

2.5.4 Ablation Study on Modality Impact

Our ablation study on modality impact, detailed in Table 2.1, reveals the effectiveness of the multimodal approach. The full model with five modalities (noisy image, noisy signal, noiseless image, noiseless signal, and noise) achieves 83.50% accuracy. Performance degrades gracefully as modalities are removed, with even the single-modality version maintaining 81.30% accuracy. This suggests that while additional modalities contribute to performance gains, the model maintains robust performance even with reduced modality input,

indicating efficient cross-modal learning capabilities.

2.5.5 Comparative Analysis

In Table 2.2, we demonstrate the superior performance of DenoMAE against existing modulation classification approaches through comprehensive empirical evaluation. Our method achieves state-of-the-art accuracy of 83.5% using only 10,000 pretraining and 1,000 fine-tuning samples, representing a dramatic reduction in data requirements compared to previous methods (e.g., 80× fewer samples than AlexNet, 10.7× fewer than NMformer). Notably, DenoMAE maintains robust performance across challenging SNR conditions, achieving 77.50% accuracy even at -10 dB SNR, while handling a broader set of 10 modulation classes compared to methods like CNN-AMC and DL-GRF which only classify 4 classes. This performance significantly outpaces existing approaches, with improvements of up to 33.1% over CNN-AMC (50.40%) and 24.5% over DL-GRF (59%) under comparable or more challenging conditions, demonstrating DenoMAE's exceptional efficiency and generalization capabilities in modulation classification tasks.

2.5.6 Cross-Framework Validation

The learned representations from DenoMAE demonstrate strong transferability across the dissertation's framework. In Chapter 3, we further validate DenoMAE's vision-based approach through RF drone detection, where representing signals as images enables successful transfer of computer vision models (YOLOv8, YOLOv11) to wireless tasks. This cross-validation confirms that the image-based signal representation and self-supervised pretraining strategy generalize beyond modulation classification. Additionally, the robust features learned by DenoMAE can serve as inputs to the statistical feature extraction pipeline in the Plug-and-Play LLM framework (Chapter 4), where pre-denoised constellation diagrams

improve cumulant computation accuracy. The encoder's ability to learn from minimal labeled data (1,000 samples) aligns with Meta-Task's few-shot adaptation paradigm (Chapter **??**), suggesting potential synergies in combining self-supervised pretraining with meta-learning for even greater data efficiency.

2.6 Conclusion

In this chapter, we introduced DenoMAE, a multimodal masked autoencoder that demonstrates the viability of transferring vision-based self-supervised learning techniques to the wireless domain. By representing wireless signals as images (constellation diagrams) and treating noise as an explicit modality, DenoMAE successfully answers **RQ1** affirmatively: vision transformers trained with masked autoencoding can effectively learn robust wireless signal representations through cross-modal knowledge transfer. Our extensive experiments validate three key findings: (1) the novel noise-as-modality paradigm enables superior denoising and robust feature learning, with 22.1% accuracy improvements at -10 dB SNR; (2) multimodal learning across five modalities (clean/noisy constellation diagrams, clean/noisy time-domain signals, and noise) provides graceful performance degradation, maintaining 81.30% accuracy even with single-modality input; (3) the approach achieves state-of-the-art performance (83.5% accuracy across 10 classes) while requiring only 10% of typical pretraining data and 33% of labeled fine-tuning data, demonstrating exceptional efficiency.

Connection to the Dissertation Framework: DenoMAE addresses the fundamental challenge of learning robust representations with minimal labeled data, providing the encoder backbone for the modular wireless foundation model framework proposed in this dissertation. Its dramatic data efficiency (10$\times$ reduction in pretraining data, 3$\times$ reduction in labeled data) validates that strong representations can be learned economically through

strategic cross-modal transfer rather than massive isolated training. This work complements the training-free reasoning capabilities of the Plug-and-Play LLM (Chapter 4), which leverages pre-trained language models to perform classification without wireless-specific training; the cross-modal validation provided by CDRF (Chapter 3), which demonstrates that vision models generalize to RF drone detection; and the rapid adaptation enabled by Meta-Task (Chapter **??**), which achieves strong performance with only 5-10 labeled examples per class. Together, these contributions demonstrate that wireless foundation models can be built through modular, cost-effective components—each approximating and integrating world knowledge from different domains—rather than requiring massive-scale end-to-end training. DenoMAE's success establishes the feasibility of the representation learning component, proving that vision-based self-supervised learning can serve as an effective pathway for economical wireless AI systems.

Chapter 3

CDRF: A Large-Scale RF Benchmark and Toolkit for Drone Detection, Classification, and Open-Set Recognition

3.1 Introduction

The development of wireless foundation models requires validating a fundamental hypothesis: can representations learned by vision models transfer effectively to the wireless domain? Chapter 2 proposes that multimodal masked autoencoders can learn universal wireless signal representations by treating signals as images (spectrograms, constellation diagrams). Before pursuing this costly self-supervised learning approach, we must first establish that vision architectures—YOLO for detection, ResNet for classification—can indeed extract meaningful features from RF signals represented as spectrograms. This chapter addresses Research Question 1 (RQ1) through a practical validation: the CageDroneRF (CDRF) benchmark demonstrates that pre-trained computer vision models successfully

transfer to wireless perception tasks, providing empirical evidence for the cross-modal transfer principle underlying our modular framework.

Beyond this theoretical validation, CDRF serves a critical practical role: providing a realistic, stress-tested benchmark for evaluating foundation model components. While many wireless datasets capture signals under idealized laboratory conditions, operational deployment demands robustness to low signal-to-noise ratios (SNR), spectrum interference, frequency offsets, and domain shifts (indoor↔outdoor). CDRF's raw-signal augmentation pipeline and diverse collection methodology create exactly these challenging conditions, enabling us to assess whether foundation model approaches generalize beyond clean benchmarks to real-world scenarios.

The rapid proliferation of Unmanned Aerial Vehicles (UAVs) provides a compelling application domain for this validation. UAVs enable transformative applications in logistics, agriculture, and emergency response [30], while introducing risks to safety, privacy, and critical infrastructure [31]. Traditional sensing modalities—radar, vision, acoustics—exhibit notable limitations: environmental sensitivity, limited small-target range, line-of-sight dependence, and high deployment costs. Radio-Frequency (RF) sensing offers complementary advantages: continuous command-and-control links enable day/night, non-line-of-sight detection at low cost, with RF captures carrying rich device, protocol, and operator signatures [32].

Despite these advantages, progress in Machine Learning (ML) for RF-based drone perception has lagged behind vision and language domains—precisely the gap motivating this dissertation's framework. A central barrier is the absence of broad, standardized benchmarks that test models under realistic stress conditions. Existing datasets are often narrow in class diversity, limited in volume, collected in clean environments with restricted SNR ranges and weak interference diversity. Models trained on such data overfit to idealized conditions, achieving near-perfect scores on "easy" benchmarks yet degrading severely in

operational deployments with Bluetooth/Wi-Fi interference, spectrum crowding, and low SNR—exactly the conditions that foundation models must handle [33].

We introduce CageDroneRF (CDRF), a benchmark built to close this gap with a dataset-and-toolkit co-design. CDRF comprises real-world captures from controlled RF-cage facilities and open-campus settings, paired with a principled raw-signal augmentation pipeline that programmatically varies SNR, injects interfering emitters, and applies frequency shifts while preserving label consistency. Crucially, augmentations operate on complex baseband I/Q, before time-frequency conversion, so synthesized conditions faithfully reflect RF phenomena rather than image-level artifacts. For detection, frequency shifts are accompanied by exact recalculation of You-Only-Look-Once (YOLO [34])-format annotations with correct wrap-around behavior on the spectrogram's frequency axis. The data processing stack converts raw captures to spectrograms via Short-Time Fourier Transform (STFT) and exposes all parameters (e.g., sampling rate, FFT size, segment length) for reproducible ablations and reprocessing.

Beyond the dataset, CDRF provides interoperable utilities for dataset creation, cleaning, and evaluation. The data module loads .dat files via memory mapping, slices long recordings into time windows, generates spectrograms, appends per-sample metadata (including time bounds), and optionally adds controlled AWGN, Rician, or Rayleigh fading noise for precise SNR targets. The YOLO toolkit includes raw-IQ augmentation and automatic bounding-box recomputation; a dataset cleaner aligns third-party releases (e.g., Roboflow-style [35]) into a consistent hierarchy; and a lightweight patch exposes per-detection class probabilities from YOLO for richer calibration and analysis. For classification, CDRF ships PyTorch [36] baselines (binary and multi-class) built on ResNet-18 [37] with ready-to-run dataloaders for both spectrogram images and array/pickle formats, standardizing training, metrics, and confusion matrices. All tooling is dataset-agnostic, enabling applications to use existing public datasets to facilitate cross-benchmark comparability.

Collectively, CDRF aims to shift the field from narrow, clean benchmarks toward realistic, stress-tested evaluation under varied SNRs, interference, and frequency offsets, conditions that operational systems routinely encounter. By releasing both data and the underlying signal-first augmentation/evaluation stack, we seek to catalyze reproducible research on RF-based detection, identification, and related perception tasks. We summarize our key contributions as follows:

- **CDRF benchmark dataset:** Real-world RF captures from cage environments with standardized spectrogram generation and rich per-sample metadata.
- **Raw-signal augmentation pipeline:** Controlled SNR injection, interfering-signal mixing, and frequency shifts applied at I/Q level, with exact YOLO label recomputation (including wrap-around).
- **SNR-structured datasets:** Programmatic creation of SNR-stratified splits (including noise-only backgrounds) for stress testing and realistic robustness evaluation.
- **Interoperable tooling:** Open-source utilities for dataset creation, metadata generation, Roboflow-style cleaning, and spectrogram rendering; tools are dataset-agnostic for use on existing releases.
- **Detection analysis enhancements:** Patch exposing per-detection class probabilities from YOLO to support calibration studies and open-set analyses.
- **Baselines and evaluation:** Ready-to-run PyTorch baselines for binary and multi-class classification with standardized metrics and confusion matrices, enabling reproducible benchmarking across tasks.
- **Open set recognition:** Implementation of open-set recognition capabilities to handle previously unseen classes during inference.

Within the broader dissertation framework, CDRF serves as a critical validation domain demonstrating cross-modal knowledge transfer. By converting RF signals to spectrogram images, this work validates that vision models (YOLO, ResNet) can transfer learned representations to wireless perception tasks—a key principle underlying the multimodal approach in DenoMAE (Chapter 2). This empirical validation provides the foundation for the dissertation's modular, cost-effective approach: rather than training wireless foundation models from scratch, we can strategically leverage pre-trained vision architectures (proven here), integrate LLM reasoning (Chapter 4), and enable rapid adaptation via meta-learning (Chapter **??**). CDRF provides the realistic, stress-tested benchmark needed to evaluate whether these foundation model components generalize beyond clean laboratory conditions to operational deployment scenarios.

The remainder of this paper is organized as follows. Section 3.2 reviews related work in RF-based drone detection, covering various sensing modalities and existing public datasets. Section 3.3 details the design and collection of the CDRF dataset, and Section 3.4 describes our data preprocessing, augmentation, and annotation pipeline. Section 3.5 presents the experimental setup and benchmark results for several machine learning tasks, including drone detection, single-label, open-set, and hierarchical classification. Finally, Section 5.5 concludes the paper and discusses future research directions.

3.2 Related Work

For a comprehensive review of drone detection methodologies, RF signal representations, deep learning approaches, and public datasets, please refer to Chapter **??**, Section **??**. This section briefly positions CDRF within the existing landscape and highlights the specific gaps it addresses.

Despite growing public datasets, persistent challenges remain: lack of diversity in device classes and environments, restricted raw I/Q access, poorly characterized SNR distributions, scarcity of real-world negative samples, and absence of integrated augmentation and evaluation frameworks. Existing datasets often achieve high in-benchmark accuracy but exhibit poor real-world generalization due to clean collection conditions [38, 39].

CDRF addresses these gaps by providing: (1) real-world cage and outdoor captures with controlled SNR stratification, (2) raw-signal augmentation pipeline operating at I/Q level with exact label recomputation, (3) dataset-agnostic tooling for spectrogram generation and evaluation, and (4) enhanced detection analysis exposing per-class probabilities for calibration and open-set studies.

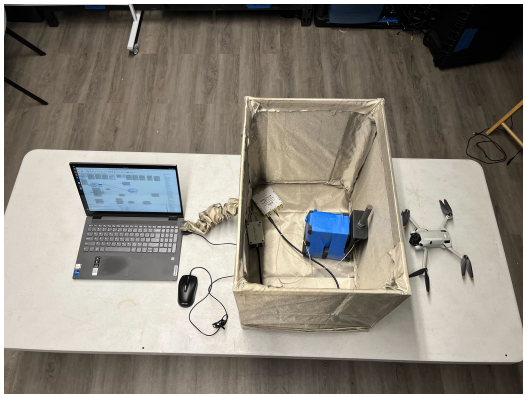
3.3 CDRF: Design and Collection

In the CDRF dataset, we aim to cover the RF signals emitted from the most common UAV-related devices in the market, including the downlink video signals and uplink Remote Controller (RC) signals. We start building the dataset by recording the UAV signals in a rather clean RF environment, in order to emphasize the characteristics of the signals of interest, and also relieve the impact of interfering RF sources or environmental variations. The clean data can also help on reducing the efforts on data-preprocessing and any following data augmentation processes.

3.3.1 Data Acquisition Platform

We built the data collection platform using a Software-Defined Radio (SDR) based system. In order to create an RF-isolation platform, we built a customized Faraday cage and placed both the SDR card and receiving antenna inside the cage to mitigate the interfering RF

(a) Recording configuration.



(b) Recording equipments.

Figure 3.1: Equipment used to capture the data: portable RF shielded enclosure, SDR Card, a laptop, and a drone.

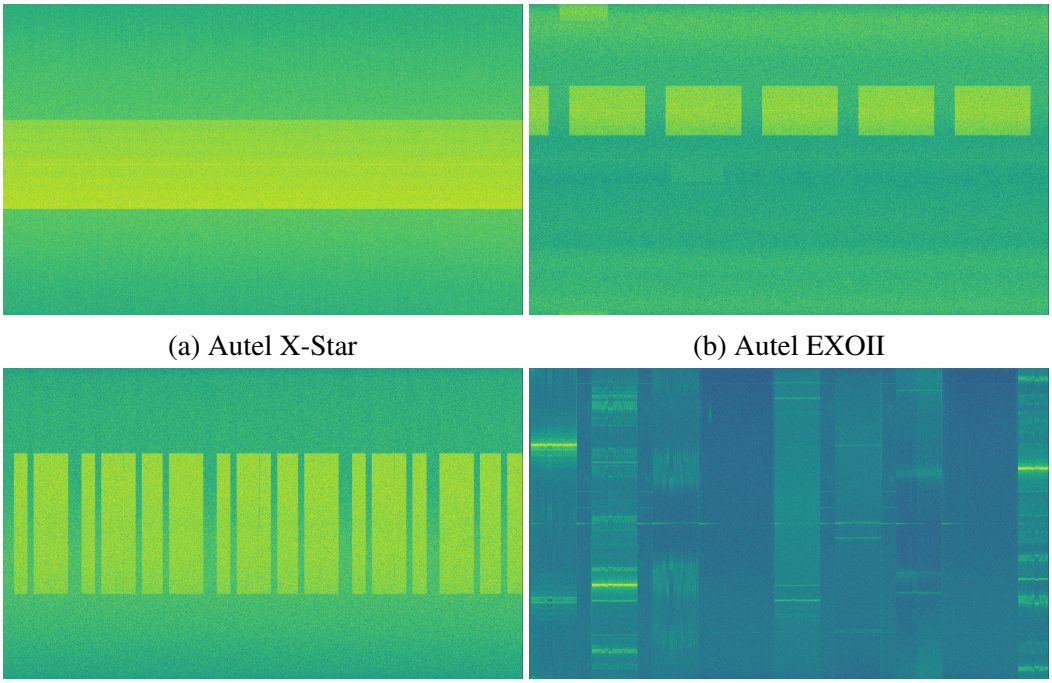
signals. The hardware setups and platform configurations are shown in Fig. 3.1.

We use a Lenovo IdeaPad Flex 5 as the data sink. The machine is equipped with an 8-core Intel Core i7-1165G7 CPU and 16 GB of memory. The operating system running on the machine is Ubuntu 24.04.2 LTS. The data collection flow and scripts are built through GNU Radio v3.10 to bridge the Universal Software Radio Peripheral (USRP) B200-mini for RF signal receiving and processing. The SDR device is then connected to a dual-band omnidirectional antenna, which is attached to a customized mount unit to ensure the antenna is static during data collection.

To ensure comprehensive coverage of drone transmissions, we employ sweep scanning across relevant frequency bands during data collection. This approach allows us to detect and record signals from devices operating at variable frequencies.

3.3.2 Data Collection Methodology

To initiate the data collection, we set up the platform as described in Fig. 3.1 (a), then turn on the UAV and the binding controller. While leaving the antenna inside the cage, we place the UAV outside and at a very close distance to the antenna. This placement ensures



(a) Autel X-Star (b) Autel EXOII

(c) DJI Mavic 2 Pro (d) RadioMaster TX16S

Figure 3.2: Representative spectrograms from the CDRF dataset, illustrating the diversity of signal types and device behaviors captured during controlled data collection.

that the UAV signals delivered to the antenna are strong enough, while other inferencing RF sources in the surrounding areas are at further distances, and thus their signals will be mostly blocked by the Faraday cage. In addition, the RC of the UAV is kept in another room next to the one where the data is collected. Our goal is to minimize the RC signals captured by the collection platform in this phase, so that the dataset can support further analysis on UAV signals and RC signals separately. This can also provide more flexibility for further data augmentation.

Next, we adjust the parameters on both the SDR devices and the UAVs to ensure they are properly aligned for optimal data recording quality. This includes settings such as the gain of the SDR device, the center frequency, and the bandwidth of the UAV's video transmission. In this work, we set the gain of the SDR device to 50 dB for indoor use and 76

dB for outdoor use, with a sampling rate of 20 MHz for both environments. For the video signal from UAVs, we select one of the channels in each of the frequency bands supported by the UAV, e.g., 900MHz, 2.4GHz, or 5.8GHz. Generally, we choose the channels that are not commonly occupied by other non-UAV wireless traffic in order to ensure the quality of the collected data. After the UAV transmission channel is defined, we tune the center frequency of the SDR device to align it to the UAV channel. This ensures that any UAV signals with a bandwidth less than or equal to 20MHz can be fully captured. In addition, this frequency alignment can help simplify the data preprocessing step for any research focusing on frequency domain representation of the signals, because the resulting transformation from the raw data, e.g., spectrogram, will contain a centralized UAV signal.

After that, we run the data recording script from the laptop for a predefined time duration, and the raw data is generated and stored in a {drone_manufacturer}_{drone_model}_{bandwidth}_{drone_center_freq}_{drone_operation_mode}.dat file along with all the predefined labels. In CDRF, we record the UAV manufacturer, model, center frequency of UAV signals, and bandwidth for each recording in the corresponding filename or directory name.

Finally, we implement additional software scripts, including a signal playback module in GNU Radio, to load the raw data file, to review the validity of the recorded raw data, and to ensure the signals recorded match the predefined parameters. We also estimate the SNR and the channel condition to make sure the recorded signals are in a rather clean state. Fig. 3.2 shows some representative spectrograms from the CDRF dataset, illustrating device behaviors captured during controlled data collection.

3.3.3 Collection of Outdoor Data

To complement our controlled indoor recordings, we conducted extensive outdoor data collection sessions. This effort was aimed at enhancing the diversity and realism of the CDRF dataset by capturing UAV signals in complex RF environments, which better reflect real-world operational conditions with various signal sources and environmental factors.

For each outdoor session, we meticulously recorded a comprehensive set of metadata to ensure the data's utility and reproducibility. The raw I/Q data files are organized into a structured directory hierarchy where each directory name systematically encodes the key parameters of the recording session. The naming convention is as follows:

`{device}_{status}_{env}_{sdr_gain}_{splitter}_{duration_recording}_{distance}_{altitude}_{center_freq}_{drone_c_freq}_{bw}_{snr}_{sampling_rate}_{record_dir}.dat`

The fields in the directory name are defined as:

- `device`: The model of the drone or device being recorded.
- `status`: The operational state of the drone (e.g., hovering, flying, on the ground).
- `env`: The environment where the recording took place.
- `sdr_gain`: The receiver gain of the SDR in dB.
- `splitter`: A flag indicating if a signal splitter was used.
- `duration_recording`: The total duration of the recording in seconds.
- `distance`: The horizontal distance from the receiver to the drone in meters.
- `altitude`: The altitude of the drone in meters.
- `center_freq`: The center frequency of the SDR receiver in MHz.

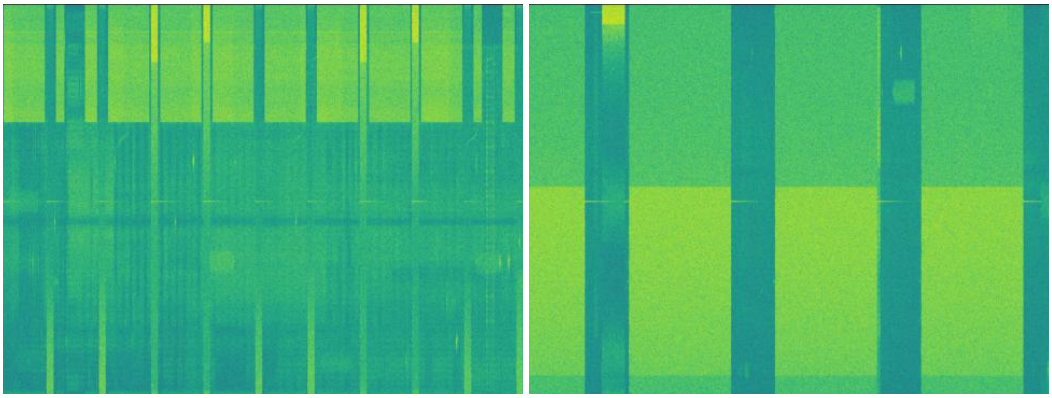
- `drone_c_freq`: The drone's transmission center frequency in MHz.
- `bw`: The signal bandwidth in MHz.
- `snr`: The estimated SNR in dB.
- `sampling_rate`: The sampling rate of the SDR in MHz.
- `record_dir`: The name of the directory where the recording is stored.

Fig. 3.3 presents a selection of representative spectrograms from our outdoor data collection, featuring signals from various drones such as the Autel EXOII, DJI Inspire1, DJI Mavic3, and DJI Phantom3 Advanced. In stark contrast to the clean, isolated signals captured indoors (as seen in Fig. 3.2), these outdoor spectrograms exhibit significantly higher levels of background noise and interference. This is visually evident from the brighter and more cluttered appearance of the spectrogram backgrounds, a direct result of the complex real-world RF environments in which the data was collected. The inclusion of this noisier, more challenging data is crucial for developing and validating drone detection models that are robust enough for practical, real-world deployment.

3.3.4 Dataset Composition and Signal Characteristics

The CDRF dataset is designed to provide a comprehensive and diverse collection of RF signals for drone detection and identification research. The dataset is built upon a large corpus of raw signal captures, totaling approximately 500+ GB of raw data.

At the core of our dataset is a significant diversity of drone models and signal types. The collection includes signals from 39 unique classes, encompassing 23 distinct commercial and hobbyist UAV models, their remote controllers, and operational variants (e.g., armed vs. unarmed states). This variety, spanning major manufacturers such as DJI, Autel, and



(a) Autel EXOII (b) DJI Inspire1

(c) DJI Mavic3 (d) DJI Phantom3 Advanced

Figure 3.3: Representative spectrograms from the outdoor dataset, showcasing the variability in signal characteristics due to environmental factors.

Parrot, ensures that models trained on CDRF are exposed to a wide range of communication protocols, a substantial leap from the handful of models found in many previous datasets. In addition to drone signals, we include a substantial number of "no-drone" recordings, capturing ambient RF noise and other common interferers like Wi-Fi signals. This enables the training of robust binary classifiers that can distinguish between the presence and absence of drone activity, a critical requirement for real-world deployment. To further challenge detection and identification algorithms, the dataset also contains recordings of multiple drones operating simultaneously in the same airspace.

All raw data is stored using the In-phase and Quadrature (I/Q) sampling method. Each sample is represented as an interleaved binary bitstream of two 32-bit floating-point numbers, capturing the full complex signal. This raw format is crucial as it allows researchers to develop and test novel signal processing and feature extraction techniques that operate directly on the baseband signal, prior to any transformations like the STFT.

The signals captured in CDRF exhibit a variety of characteristics typical of modern drone communication systems. Many of the drones utilize Frequency-Hopping Spread Spectrum (FHSS) for their command and control links. These signals are defined by key parameters such as their hopping frequency, duration, duty cycle, and the period of their hopping pattern. In contrast, signals used for video transmission typically have wider bandwidth and longer duration than control signals. We define the collection of these characteristics as the RF Drone Fingerprint, which serves as a rich input for deep learning models to differentiate between drone types.

Furthermore, our data captures the inherent behavioral variability of drone signals. For instance, the frequency hopping patterns of a drone during its initial pairing process can differ significantly from those during normal flight operations. The dataset also captures more subtle phenomena, such as the variable time duration of video transmission signals from certain drone models, a characteristic that can serve as another important feature for robust classification.

3.4 Data Preprocessing, Augmentation, and Annotation

We release both a raw-signal processing pipeline and augmentation utilities designed to create realistic, label-consistent training data from complex I/Q captures. All components are open-source and parameterized for reproducibility.

3.4.1 Spectrogram Generation

Raw complex I/Q streams are transformed into time–frequency representations via the STFT. Given a complex signal $x[n]$, window $w[n]$, window length N, and hop size H, the STFT is [40]

$$X(m,k) \;=\; \sum_{n=0}^{N-1} x[n+mH]\; w[n]\; e^{-j2\pi kn/N},$$

where m is the discrete time index and k is the discrete frequency index.

We compute the power spectrogram $S(m,k) = |X(m,k)|^2$ and render in dB as

$$S_{\mathrm{dB}}(m,k) \;=\; 10\log_{10}\big(S(m,k)+\epsilon\big),$$

where ϵ represents a very small, positive constant used to avoid taking the logarithm of zero. After that map to RGB using a perceptual colormap. By default, we use `scipy.signal.stft` [41] with a Hann window, $N =$ `FFT_SIZE` $= 1024$, overlap $= 128$ samples, and return two-sided spectra with DC centered via `fftshift`. Default sampling is `SAMPLING_RATE` $=$ 20 MHz and we generate `NUM_FFT_SPEC` $= 1500$ time bins per sample. Spectrograms are exported as PNG images alongside per-sample metadata.

3.4.2 Frequency Resolution and Colormap Selection

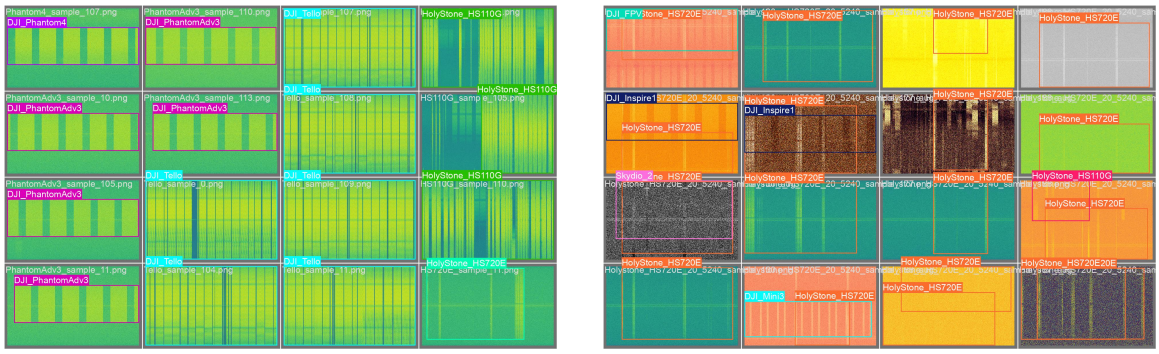
The window size N and hop ($N -$ overlap) set the time–frequency trade-off; larger N improves frequency resolution at the cost of temporal precision. All parameters are exposed to support ablations. We provide multiple colormaps (`viridis`, `plasma`, `inferno`, `magma`, `cividis`, `gray`, `hot`) to test color sensitivity; the default pipeline normalizes spectrograms to $[0,1]$ before colorization and composites to RGB for consistent rendering.

3.4.3 Data Normalization

We include signal-level and spectrogram-level normalization operators:

- *Power normalization* of complex I/Q to unit average power: $\tilde{x} = x/\sqrt{\mathbb{E}|x|^2 + \epsilon}$.
- *Z-normalization* of real and imaginary parts independently to zero mean, unit variance.
- *Spectrogram normalization* options: per-frequency z-score, per-time z-score, or global min–max to $[0, 1]$ prior to colorization.

These are selectable in the augmentation scripts and can be toggled per experiment.



(a) Annotated Data. (b) Augmented Data.

Figure 3.4: Examples of annotated and augmented data.

3.4.4 Custom Augmentation Tools

Our augmentation operates at the *raw-signal* (I/Q) level to better preserve RF physics and then projects to spectrograms, with exact label propagation to detection annotations.

3.4.4.1 SNR Conditioning via AWGN

To synthesize controlled SNRs, we add complex AWGN $n \sim \mathcal{CN}(0, \sigma^2)$ to x where the noise power σ^2 is chosen for target $\mathrm{SNR}_{\mathrm{dB}}$:

$$P_s = \mathbb{E}|x|^2, \quad \mathrm{SNR}_{\mathrm{lin}} = 10^{\mathrm{SNR}_{\mathrm{dB}}/10}, \quad \sigma^2 = P_s/\mathrm{SNR}_{\mathrm{lin}}.$$

We optionally export noise-only spectrograms for analysis and create SNR-stratified dataset variants (e.g., per-SNR subdirectories), enabling robustness sweeps.

3.4.4.2 Frequency Shifting

We simulate carrier offsets and front-end errors by multiplying with a complex exponential

$$x_{\Delta f}[n] \;=\; x[n]\, e^{j2\pi\Delta f\, n/F_s},$$

which induces a vertical translation on the spectrogram and where F_s is sampling rate. This is applied before STFT and is label-consistent with our detection annotation recomputation (below).

3.4.4.3 Interferer Mixing

To emulate spectrum crowding, we mix two normalized signals at a controllable ratio $\alpha \in [0, 1]$:

$$x_{\text{mix}} \;=\; \mathrm{norm}\left(x_1 \;+\; \alpha\, x_2\right),$$

optionally with independent frequency shifts on each source. This produces realistic co-channel scenarios directly in the I/Q domain.

3.4.4.4 Spectrogram Rendering Variants

Augmented samples are rendered under randomized, but reproducible, choices of colormap and normalization policy to increase visual diversity while keeping frequency content intact.

3.4.4.5 Background Noise and Non-Target Signals

Our dataset includes a comprehensive corpus of non-target signals, captured from diverse environments where no drones were active. To facilitate common machine learning workflows, we provide a suite of tools to extract these "no-drone" segments and generate unified metadata files for mixed-signal experiments. This curated collection of real-world negative samples is crucial for training well-calibrated binary detectors and is explicitly designed to support advanced strategies like hard-negative mining.

3.4.5 Annotation Strategy

We adopt YOLO-format annotations on spectrogram images with a *whole-signal* policy: contiguous RF emission bursts from a device are annotated as a single object to capture both spectral morphology and temporal occupancy. Fig. 3.4 shows examples of annotated data and augmented data with annotations.

In general, a spectrogram of a UAV signal segment consists of a sequence of ON-OFF states, which represents the transmission pattern of the UAV in both time and frequency domain. In order to lead YOLO model to better capture the signal characteristics in both domains, we define the annotation policy as that, the bounding box for each type of signals in a spectogram image will start and end with an ON state of the signal. The OFF states at the beginning and end of a spectrogram are neglected. This can help the YOLO model to track the start and end of a transmission, and thus to output more accurate bounding

boxes. Meanwhile, if the RF emission bursts captured by a spectrogram are with limited time duration, i.e., $< 10\%$ of the time duration of the spectrogram, or there is only a single ON state in the whole spectrogram, no bounding box annotation will be given, and the spectrogram image will be labeled as background in those cases.

As for the RC signals, due to its frequency-hopping property, in CDRF we aim at providing two types of YOLO-formated annotation to support the flexibility of future research or implementation in potentially different purposes.

- We provide one bounding box covering all the RC signal bursts captured in a spectrogram. Similar to the UAV signal labeling, we ensure that the edges of the bound box align with the start and end of the sequence of RC signal bursts

- CDRF also contains annotation for RC signal bursts transmitted at the same channel. Since the frequency-hopping behavior is not guaranteed to occur or being captured in every single spectrogram in a continuous raw data recording, we believe that this type of annotation can contribute the learning of YOLO models and improve the detection performance when deploying it in reality.

Moreover, due to the fact that most of the RC signals are narrowband and in short duration for each ON state, we do not provide any bounding box annotation to the spectrogram that contains only one single ON state. The reason behind this design choice for annotation strategy is that the bound box covering only single ON state does not provide any information for the model to learing about its pattern across either time or frequency domain. Similar statement on UAV video signal annotation is also demonstrated in [42]. Also, the detection or classification based on a single ON state is generally neither expected nor considered reliable in real-world UAV detector deployment scenarios.

3.4.5.1 Label-Preserving Transforms with Wrap-Around

Frequency shifts correspond to vertical translations in normalized image coordinates by $\Delta y = \Delta f / F_s$. We recompute bounding boxes analytically after augmentation and correctly handle wrap-around at the top/bottom edges: if a box straddles the boundary after shift, it is split into two valid boxes whose heights sum to the original. Boxes are also filtered by a configurable minimum height to avoid degenerate labels after extreme shifts.

3.4.5.2 Dataset Hygiene and Third-Party Integration

We include utilities to normalize third-party releases (e.g., Roboflow-style layouts) into a consistent hierarchy (`images/ labels/` with class folders and {train,val,test} splits) and to preserve class mappings. This enables direct augmentation and training across heterogeneous sources.

3.4.6 Reproducibility and Metadata

Each sample carries rich metadata to ensure reproducibility: original file path, device/model label, sampling rate, FFT size, number of STFT frames, per-sample time bounds (start/end in seconds), center frequency, and output paths for spectrograms. All processing parameters (e.g., window size, overlap, SNR target, shift amounts, mixing weights) are saved or deterministically re-creatable from configuration, facilitating exact regeneration of training splits and ablation studies.

3.5 Machine Learning Tasks and Baselines

This section details the machine learning tasks supported by our dataset, the baseline models implemented, and their performance. We address drone detection, single-label classification, multi-label classification, open-set recognition, and hierarchical classification.

3.5.1 Drone Detection (YOLO)

3.5.1.1 Task Definition

Given a time–frequency spectrogram, the goal is to detect and localize RF emissions attributable to drones. The detector predicts a set of axis-aligned bounding boxes with class labels in YOLO normalized coordinates, where each object corresponds to a contiguous emission burst (whole-signal policy) on the spectrogram.

3.5.1.2 Baseline Model Architecture and Training

We use a single-stage object detector, `YOLOv11n` [43] (Ultralytics [44] v8.3.156), trained on spectrograms produced by our pipeline (section 3.4). We evaluate two scenarios: (i) Clean, in which spectrograms are generated without augmentation; and (ii) Augmented, in which raw complex I/Q are transformed before STFT via frequency shifting, interferer mixing, and SNR conditioning. Models are initialized from COCO-pretrained weights [45, 44, 43], trained with deterministic seeds, and validated at each epoch. Labels follow the whole-signal policy, and for frequency shifts we recompute YOLO annotations exactly with wrap-around handling on the frequency axis to preserve label consistency. Unless otherwise specified, Ultralytics defaults are used for input size, data augmentation, and non-max suppression.

3.5.1.3 Evaluation Metrics

We report standard detection metrics: class-aggregated Precision and Recall; mean Average Precision at IoU 0.5 (mAP@0.5); and COCO-style mAP averaged over IoU thresholds from 0.5 to 0.95 in steps of 0.05 (mAP@[.5:.95]). For qualitative error analysis, we also include normalized confusion matrices computed from matched detections.

3.5.1.4 Experimental Results and Baseline Performance

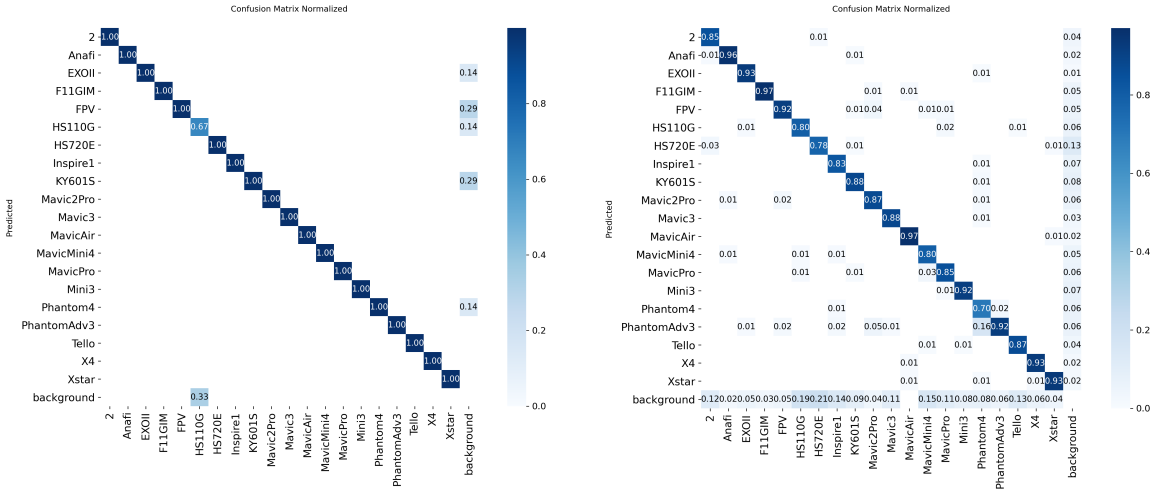
Table 3.1: YOLO detection results.

Run	Epochs	Precision	Recall	mAP@0.5	mAP@[.5:.95]
Clean	100	0.932	0.982	0.986	0.948
Augmented	50	0.940	0.855	0.935	0.838

Table 3.1 summarizes the best validation checkpoints: The model trained without any augmentations (referred to as "clean") achieves strong performance (Precision $= 0.932$, Recall $= 0.982$, mAP@0.5 $= 0.986$, mAP@[.5:.95] $= 0.948$), with the top mAP@[.5:.95] observed near epoch 78. Under augmentation, precision is maintained and slightly improved (Precision $= 0.940$) while recall drops (Recall $= 0.855$), yielding lower top-line mAP (mAP@0.5 $= 0.935$, mAP@[.5:.95] $= 0.838$; best near epoch 49). This reflects the expected precision–recall trade-off when training on harder, low-SNR and interferer-rich conditions: the detector becomes more conservative (fewer false positives) at the expense of increased misses (more false negatives).

Despite this shift, the normalized confusion matrices in Fig. 3.5 remain strongly diagonal for both regimes, indicating that whole-signal annotations preserve class separability even under frequency shifts and spectrum crowding. The dominant degradation under augmentation is recall (missed detections) rather than systematic label confusion, suggesting

room for robustness gains via longer training, class-balanced sampling, or stronger backbones without altering the annotation policy.



(a) Clean: normalized confusion.

(b) Augmented: normalized confusion.

Figure 3.5: Class-wise performance and error structure on validation.

3.5.2 Hierarchical Classification

3.5.2.1 Task Definition

To provide more robust and interpretable classifications, we explored a hierarchical approach that classifies signals in a coarse-to-fine manner. In Fig. 3.6, we provide the full structural tree of the hierarchical classification. Our hierarchy is structured into three levels: Modulation (3 classes), Protocol (5 classes), and finally the specific drone Model (27 classes).

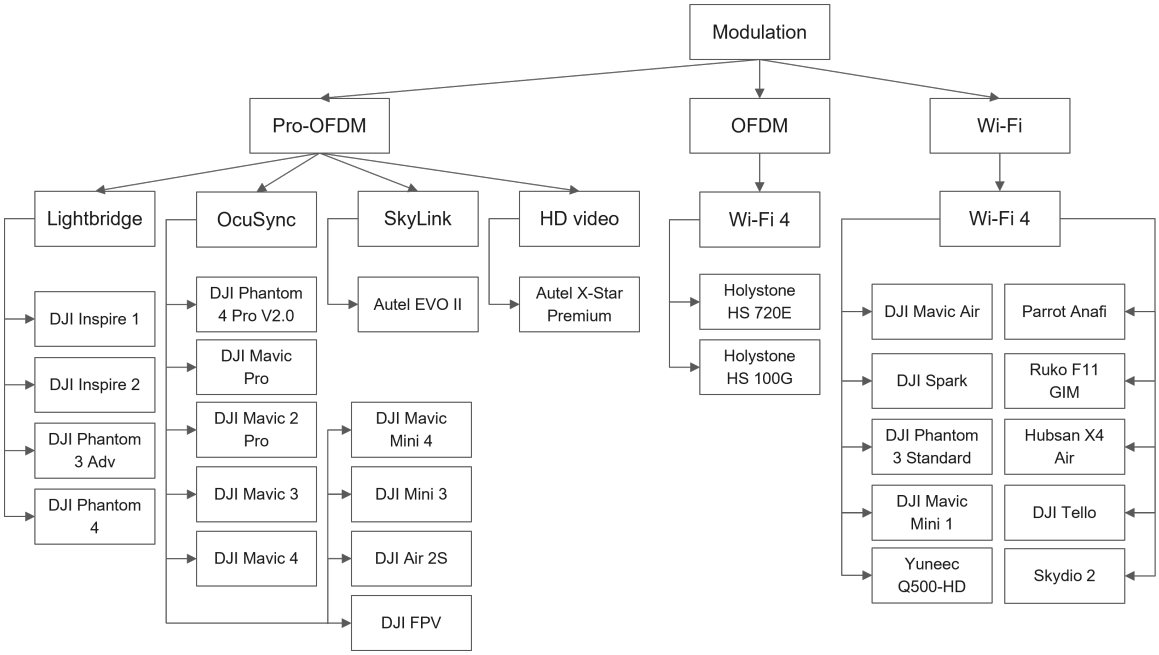


Figure 3.6: Hierarchical classification tree structure of CageDroneRF.

3.5.2.2 Baseline Model Architecture and Training

We implemented a multi-task learning model using a shared ResNet-18 [37] backbone to extract features from the input spectrograms. The output of the feature extractor is fed into three separate classification heads, each dedicated to one level of the hierarchy. The model is trained end-to-end using a "HierarchicalLoss", which is a weighted sum of the cross-entropy losses from each head.

To observe the effectiveness of our indoor data and how it generalizes to outdoor scenarios, we trained our hierarchical classifier on the indoor subset of CDRF and evaluated it on the outdoor subset. Furthermore, we examined how including outdoor data in the training set affects performance by training another model on the combined indoor and outdoor data. In both cases, the amount of indoor training data was kept constant to ensure a fair comparison. The models face challenging scenarios, including varying SNR levels, signal bandwidths, outdoor drones flying at different altitudes, distances, and the presence

of various interferers.

3.5.2.3 Experimental Results and Baseline Performance

The results, shown in Fig. 3.7, compare the generalization performance of two training configurations when tested on outdoor data across three classification levels: Modulation, Protocol, and Model.

When the model is trained on indoor data only, it achieves moderate accuracy on lower-level features (Modulation: 69.95%, Protocol: 66.11%) but struggles significantly with identifying the specific drone model (42.07%). This indicates poor domain generalization, as the model fails to adapt from indoor to outdoor environments.

In contrast, training on a combined set of indoor and outdoor data yields substantial improvements across all hierarchical levels. Accuracy increases to 89.42% for Modulation, 87.34% for Protocol, and 69.63% for Model. The most significant gain of +27.56% is observed in model-level classification, the task most sensitive to environmental shifts. This demonstrates that training on diverse, mixed-domain data is crucial for learning robust, invariant features.

3.5.3 Open-Set Recognition

3.5.3.1 Task Definition

Open-Set Recognition (OSR) extends classification to simultaneously: (1) accurately classify known drone types from the training set, and (2) identify and reject unknown samples as “novel” or “outlier” classes, preventing forced misclassification into known categories.

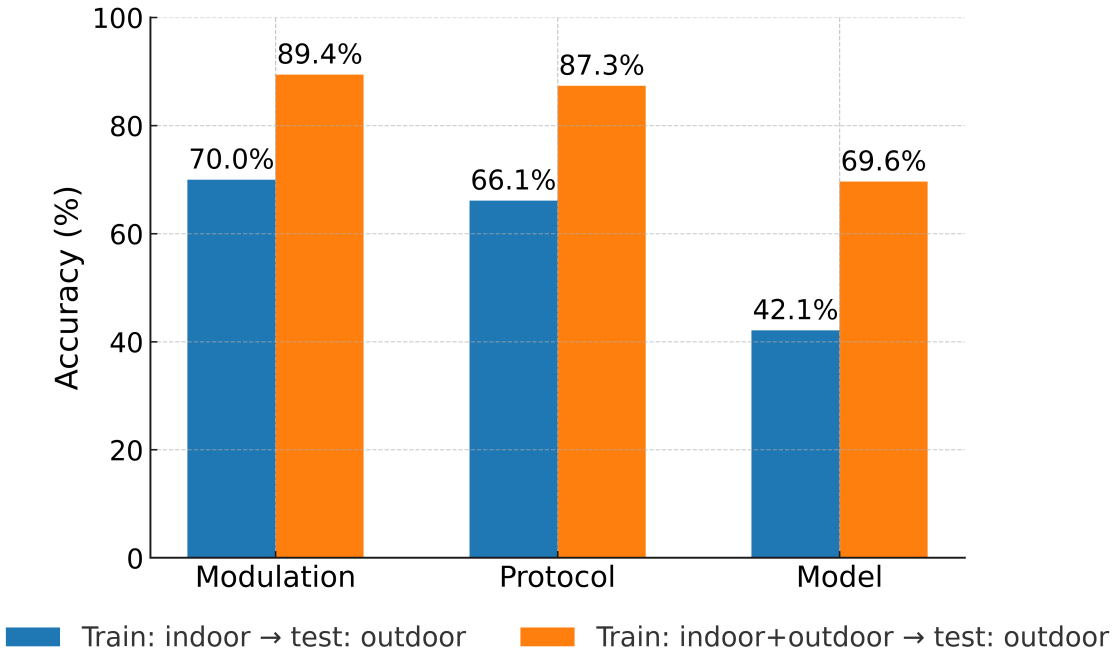


Figure 3.7: Hierarchical classification accuracy on different training setups.

3.5.3.2 Baseline Model Architecture and Training

In our experiments, we utilized MetaMax [46], a state-of-the-art OSR method that combines deep feature learning with statistical modeling of known class distributions. For our OSR experiments, we curated a subset of CDRF comprising 14 distinct drone types totaling 2,180 spectrogram images. The dataset was strategically partitioned into known and unknown classes to simulate realistic deployment scenarios. The known class set contains 10 drone types with 1,915 images, and the unknown class set comprises 4 drone types with 265 images. The data was split into training (60%) and validation (20%) samples from known classes only, with the remaining samples reserved for testing.

3.5.3.3 Evaluation Metrics

Performance metrics included: (1) closed-set accuracy on known classes, (2) unknown detection rate, and (3) overall accuracy.

3.5.3.4 Experimental Results and Baseline Performance

We evaluated MetaMax on our curated CDRF subset, achieving an overall accuracy of 75.15%, with known class accuracy of 75.98% and an unknown detection rate of 73.96%. The detailed confusion matrix (Fig. 3.8) reveals strong performance correlation with class size. Among the 265 unknown test samples, 196 (73.96%) were correctly identified as novel. The binary confusion matrix for known vs. unknown discrimination is shown in Fig. 3.9. These results demonstrate both the promise and challenges of OSR for RF drone detection.

3.5.4 Drone Classification (Single-Label)

3.5.4.1 Task Definition

The single-label drone classification task involves assigning each detected RF signal to exactly one drone type from a predefined set of classes.

3.5.4.2 Baseline Model Architecture and Training

We implemented a ResNet-18 convolutional neural network as the backbone, leveraging its proven effectiveness for spectrogram-based RF signal classification. The model was trained on spectrogram images generated from the CDRF dataset, using standard data augmentation and normalization techniques to improve generalization. We had a total number of 20 different drone types in our dataset. The ResNet-18 model was trained with a cross-entropy

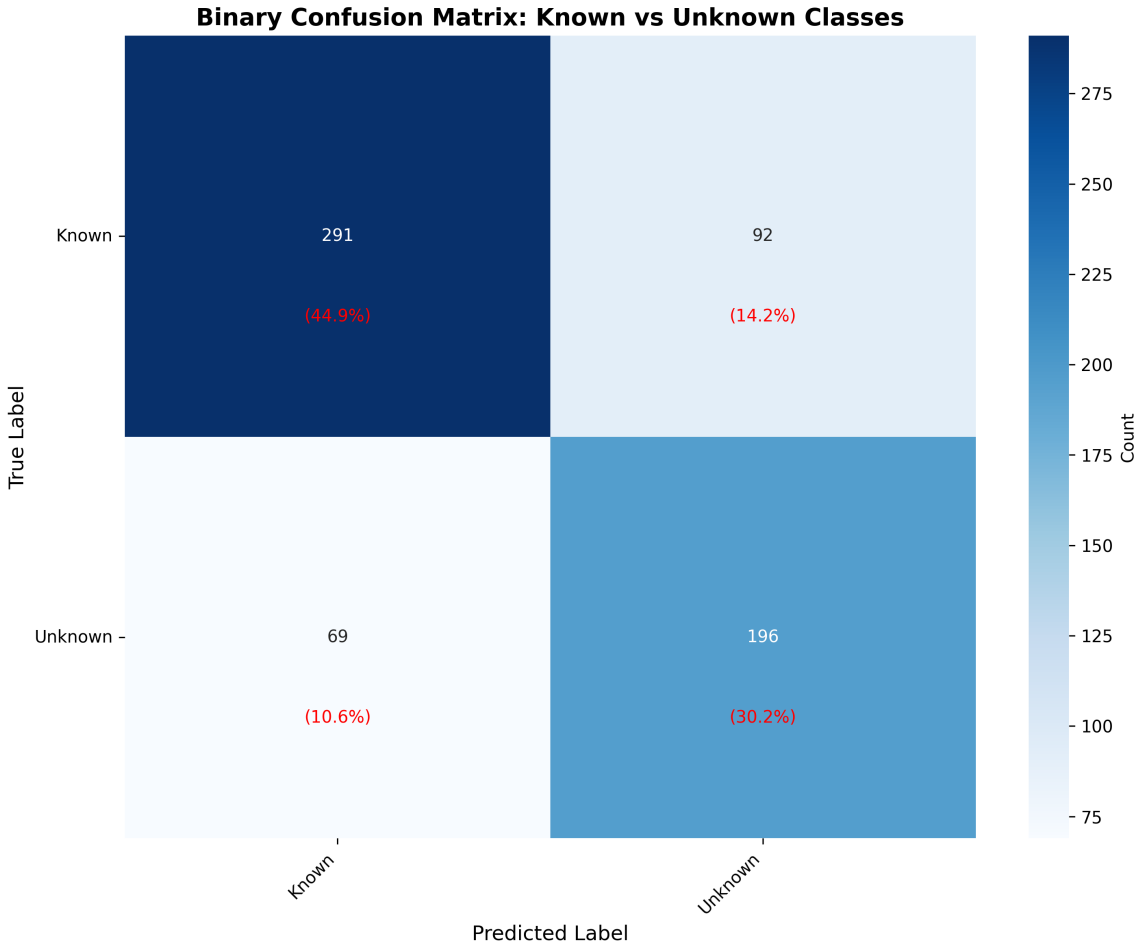


Figure 3.8: Binary confusion matrix for known vs unknown class discrimination.

loss and optimized using Adam.

3.5.4.3 Experimental Results and Baseline Performance

After training, the model achieved a test accuracy of 53.49%, as shown in the confusion matrix (Fig. 3.10). The confusion matrix reveals that certain drone types are classified with high accuracy, while others exhibit significant misclassifications, especially for classes that seen less training data, indicating areas for potential improvement. The results highlight the

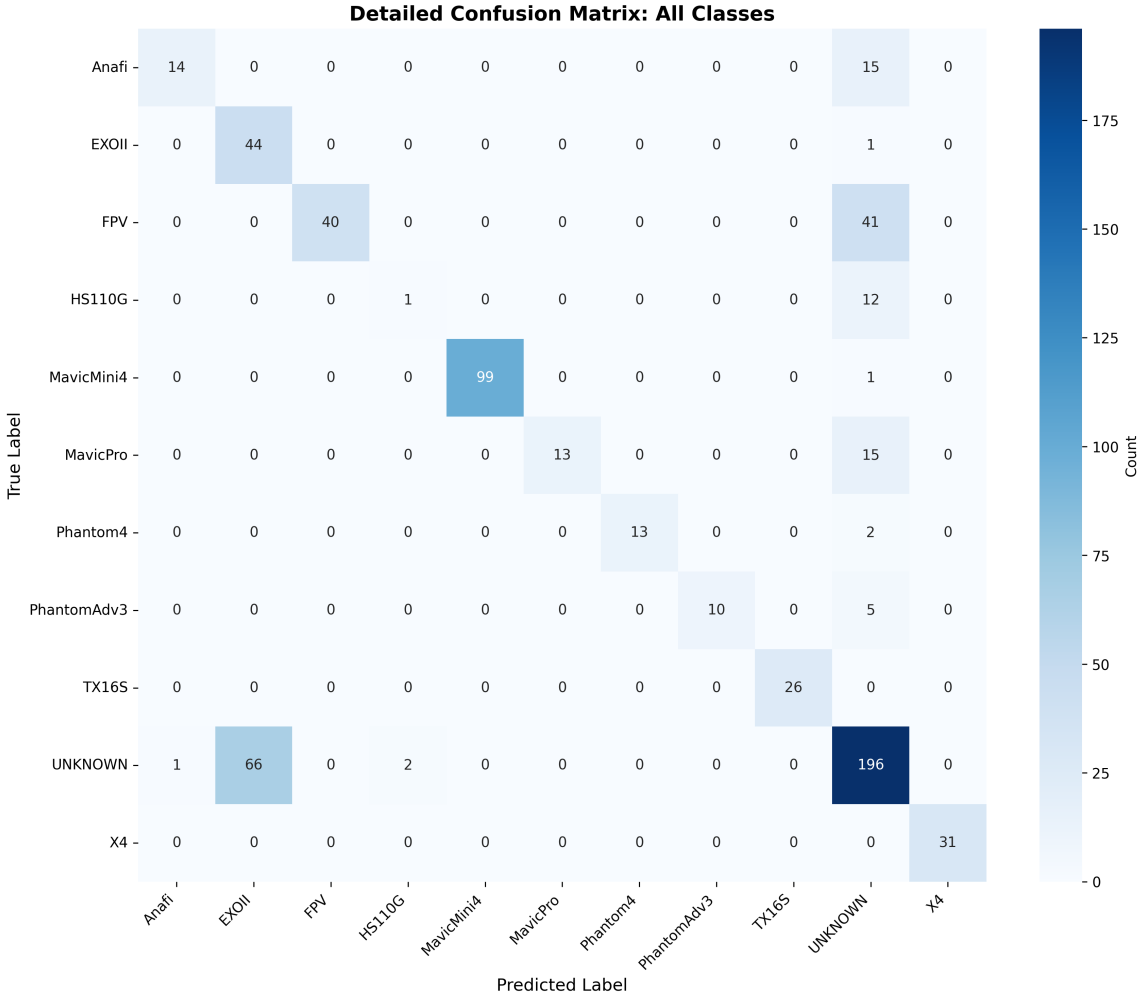


Figure 3.9: Detailed confusion matrix showing per-class performance for open-set recognition.

challenges of single-label classification in RF drone signals, particularly in distinguishing between similar models.

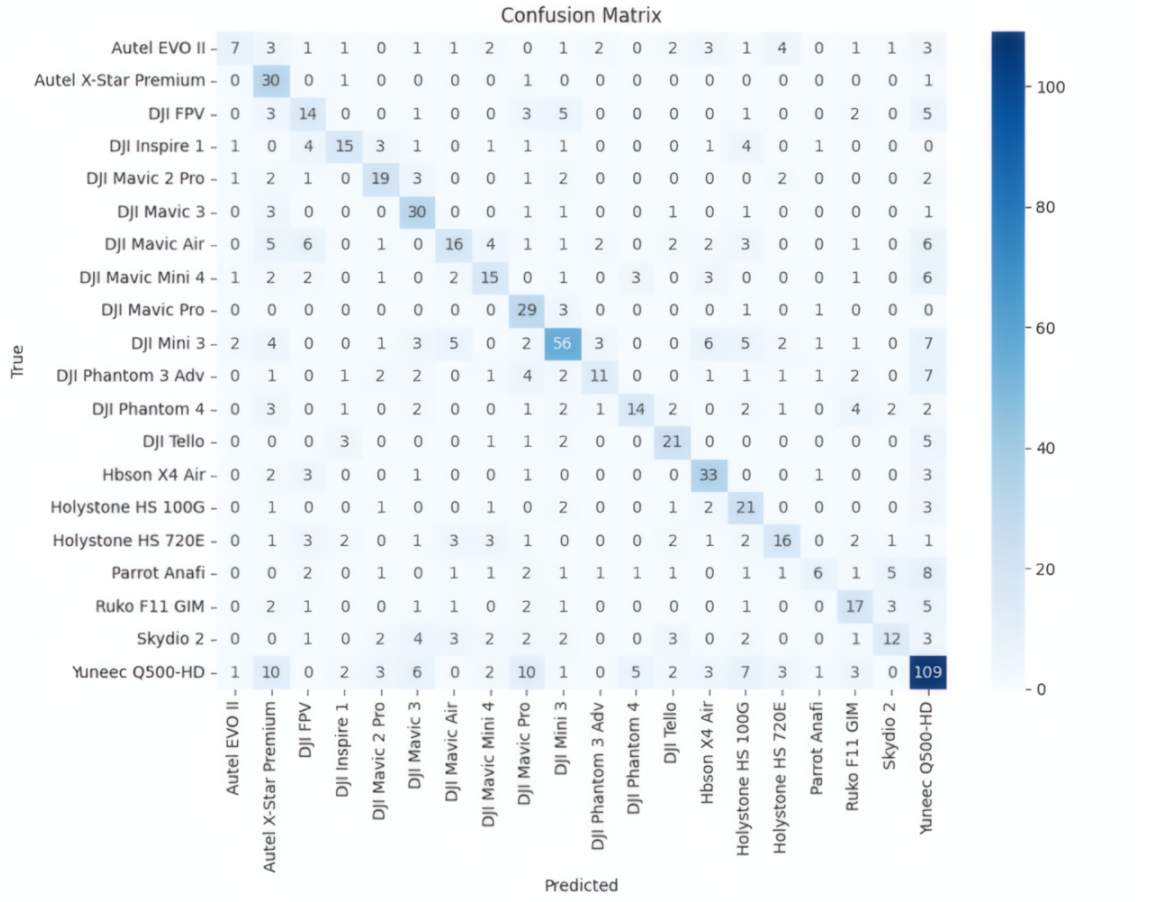


Figure 3.10: Confusion matrix for single-label drone classification.

3.6 Discussion and Future Work

Cross-Modal Transfer Validation: CDRF's results validate the central hypothesis underlying the dissertation's framework: vision models successfully transfer to wireless domains when signals are represented as images. The YOLOv11n detector achieves mAP@0.5 = 0.986 on clean spectrograms and 0.935 under augmentation, demonstrating that object detection architectures—pretrained on natural images (COCO)—extract meaningful features from RF time-frequency representations without wireless-specific modifications. Similarly, ResNet-18 baselines achieve 53.49% single-label accuracy across 20 drone types

and 75.98% known-class accuracy in open-set scenarios, confirming that convolutional architectures transfer across modalities. This empirical evidence directly supports Research Question 1 (RQ1) and justifies DenoMAE's approach of leveraging vision transformers for wireless representation learning (Chapter 2).

Robustness Gap and Foundation Model Implications: The performance degradation under augmented conditions reveals critical challenges for foundation models. The recall drop from 0.982 to 0.855 (YOLOv11n) reflects the precision-recall tradeoff when training on low-SNR, high-interference conditions—the detector becomes more conservative to avoid false positives but misses more true detections. This robustness gap motivates the dissertation's modular framework: self-supervised pretraining (DenoMAE) on diverse unlabeled signals should learn noise-invariant representations; LLM-based reasoning (Plug-and-Play) should provide complementary decision-making that doesn't degrade under noise; and meta-learning (Meta-Task) should enable rapid adaptation to deployment-specific SNR distributions with minimal labeled data. The hierarchical classification results (indoor-only: 42.07% model accuracy; mixed training: 69.63%, +27.56%) demonstrate that domain shift—a core challenge for foundation models—can be addressed through diverse pretraining data, validating the framework's emphasis on broad data coverage.

Open-Set Recognition and Real-World Deployment: The 73.96% unknown detection rate establishes that models can learn to reject novel threats, but the class-size correlation in performance (Fig. 3.8) highlights a persistent challenge: data imbalance. This finding connects directly to Meta-Task (Chapter **??**), which addresses few-shot learning scenarios where new drone types emerge with only 5-10 labeled examples. The open-set capability validated here provides a safety mechanism for foundation model deployment—rather than forcing every input into a known category, models can flag uncertainty and request human verification.

To ensure the long-term relevance and utility of CDRF, we are committed to its continuous expansion. Future work will focus on several key areas:

- **Dataset Expansion:** We will regularly update the dataset with new and emerging drone models, as well as more diverse background and interference signals captured from a wider range of environments (e.g., urban, industrial, and rural). We also plan to integrate CDRF with other public datasets to create an even more comprehensive and diverse benchmark.
- **Advanced Model Architectures:** We will explore the application of more sophisticated deep learning models, such as Transformers and graph neural networks, which may be better suited to capturing the complex temporal and spectral dependencies in RF signals.
- **Enhanced Augmentation Techniques:** We will continue to develop more advanced data augmentation techniques that can more realistically simulate a wider range of real-world channel effects, such as multipath fading and Doppler shifts.
- **Focus on Advanced Tasks:** We will further investigate the more challenging tasks of multi-label and open-set classification, which are critical for real-world applications where multiple drones may be present, and novel threats are a constant concern.

3.7 Conclusion

This chapter presented CDRF, a comprehensive benchmark that validates a fundamental premise of the dissertation's framework: vision models can successfully transfer to wireless perception tasks. By demonstrating that YOLO detection (mAP@0.5 = 0.986 clean, 0.935 augmented) and ResNet classification operate effectively on RF spectrograms, CDRF

provides empirical evidence for the cross-modal transfer principle underlying Research Question 1 (RQ1). This validation justifies the multimodal approach pursued in DenoMAE (Chapter 2), where masked autoencoders learn universal wireless representations by treating signals as images.

Beyond theoretical validation, CDRF establishes a realistic stress-test benchmark for foundation model components. The performance degradation under augmented conditions—particularly the recall drop from 0.982 to 0.855 in YOLO detection—reveals the robustness challenges that operational systems face. The hierarchical classification results (+27.56% model-level gain with mixed indoor/outdoor training) demonstrate the domain shift problem that foundation models must address through diverse pretraining and effective adaptation mechanisms. The open-set recognition capability (73.96% unknown detection) validates that models can learn to reject novel threats, a critical requirement for real-world deployment.

Framework Integration: Within the dissertation's modular system, CDRF serves three essential roles. First, it validates the feasibility of vision-to-wireless transfer, establishing that pre-trained computer vision architectures provide a strong foundation for wireless perception. Second, it provides a diverse, challenging benchmark (39 classes, 500GB raw data, controlled SNR stratification) for evaluating whether foundation model approaches—self-supervised learning (DenoMAE), LLM-based reasoning (Plug-and-Play), meta-learning adaptation (Meta-Task)—generalize beyond clean laboratory conditions. Third, its raw-signal augmentation pipeline and dataset-agnostic tooling enable systematic robustness evaluation across the SNR, interference, and domain shift conditions that real deployments encounter.

The success of vision models on CDRF's drone detection tasks demonstrates that cross-modal knowledge transfer is not merely theoretical but practically achievable. This empirical validation provides confidence for pursuing the more sophisticated approaches in

subsequent chapters: DenoMAE's self-supervised representation learning can leverage vision transformer architectures because those architectures demonstrably work for wireless signals; the Plug-and-Play framework's LLM integration becomes more plausible given that one domain (vision) has already successfully transferred to wireless; and Meta-Task's few-shot adaptation addresses the remaining challenge of rapidly specializing foundation model capabilities to new wireless scenarios with minimal data.

By open-sourcing both dataset and toolkit, CDRF aims to foster reproducible research and collaborative progress toward robust, field-ready wireless AI systems. Future work will expand the dataset with emerging drone models, explore advanced architectures (transformers, graph neural networks), and investigate how CDRF can serve as a pretraining corpus or evaluation benchmark for the integrated foundation model framework proposed in this dissertation.

Chapter 4

Plug-and-Play Alignment for Large Language Models

4.1 Introduction

Automatic Modulation Classification (AMC) is a critical technology in modern wireless communications, underpinning applications in cognitive radio, spectrum sensing, and interference management. By accurately identifying modulation schemes, AMC facilitates efficient spectrum utilization and enhances the adaptability and reliability of communication networks. However, despite its significance, AMC remains a challenging problem due to the complex interplay of signals with ambient noise and other channel impairments.

Early approaches to AMC predominantly relied on traditional machine learning techniques and handcrafted feature extraction, but these methods have gradually been superseded by deep learning models. For instance, Convolutional Neural Networks (CNNs) have been extensively applied in this domain. Peng et al. [24] transformed raw modulated signals into constellation diagrams to serve as inputs for CNN architectures such as AlexNet,

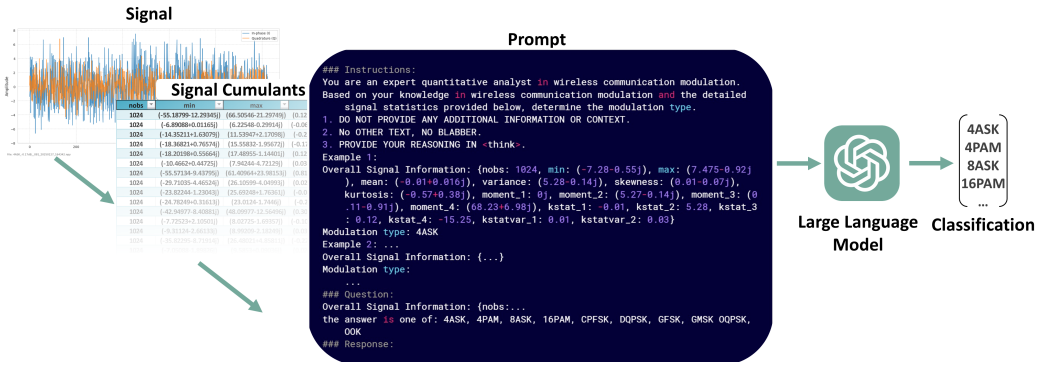


Figure 4.1: Overview of our in-context AMC framework. The raw signal is first summarized into high-order statistics and cumulants, which are then embedded in a text prompt. This prompt is fed into a large language model to perform modulation classification from a predefined set of modulation types (e.g., 4ASK, 4PAM, 8ASK, 16PAM).

while subsequent studies introduced variants like MCNet [47] that leveraged multiple convolutional blocks with skip connections and asymmetric kernels to capture spatio-temporal correlations. In another investigation, a CNN framework was explored using multiple signal representations, such as constellation diagrams, ambiguity functions, and eye diagrams, with empirical results indicating that combining these representations could improve classification accuracy [48].

More recently, attention-based models, particularly transformers, have garnered significant interest for AMC. Transformers, introduced by Vaswani et al. [49], are adept at processing sequential data by dynamically focusing on the most relevant parts of the input. Cai et al. [50] demonstrated that a transformer network (TRN)-based approach can capture long-range dependencies more effectively than conventional CNNs, which is especially beneficial in low SNR scenarios. Building on this idea, Kong et al. [8] further improved transformer-based AMC by incorporating convolutional embedding and attention pooling techniques, and Qu et al. [51] combined transformers with Long Short-Term Memory (LSTM) networks to address long-range temporal dependencies and enhance robustness against noise through novel data augmentation strategies.

Despite these advancements, many existing methods are developed under idealized, noise-free conditions [27, 52]. Such conditions are rarely encountered in real-world environments where signals are typically contaminated with noise. Moreover, several approaches struggle with fine-tuning downstream tasks when only limited labeled data is available [53, 54].

Nmformer [55] employs a Vision Transformer (ViT) trained on a large labeled dataset to achieve strong performance when retrained on low-SNR signals. However, the model struggles to attain high accuracy due to the absence of unsupervised pretraining. In contrast, DenoMAE [56] and DenoMAE2.0 [57] aim to achieve competitive performance with only a few thousand samples during pretraining and significantly fewer labeled samples in the downstream task. Nevertheless, the limited data constrains the model's ability to achieve substantial performance improvements. This underscores the need for a more advanced framework that effectively leverages large-scale data to enhance accuracy.

Addressing these limitations, recent work has begun to explore more sophisticated frameworks that can balance performance across varying SNR conditions. Gao et al. in the MoE-AMC study [7] proposed a Mixture-of-Experts (MoE) approach that integrates a ResNet-based model for high SNR signals with a transformer-based model for low SNR signals, achieving a more balanced and robust classification performance. Complementing this, the review by Jafarigol et al. [58] provided a comprehensive analysis of contemporary AI/ML-based AMC models, highlighting both recent advances and persistent challenges, particularly the difficulty of maintaining accuracy in noisy environments and the need for efficient models that operate under constrained computational resources.

In this work, we take a novel direction by leveraging a pretrained LLM-based approach for automatic modulation classification. Our method utilizes traditional modulation statistics and cumulants, capitalizing on the LLM's inherent familiarity with classical signal

processing techniques. This capability enables our framework to perform competitive one-shot modulation classification without the need for exhaustive training across all SNRs, modulation types, or channel conditions, and without the computational overhead of extra preprocessing steps such as denoising. By transforming quantitative signal features into natural language descriptions, our approach bridges the gap between classical signal processing and modern AI, achieving robust performance even in challenging, noisy scenarios.

Our contributions include:

- A novel in-context approach for AMC using LLMs that simplifies the processing pipeline.
- A one-shot classification method that eliminates the need for extensive training data.
- Competitive performance under both noiseless and noisy conditions.

Framework Position: Within the dissertation's modular framework for Wireless Physical-Layer Foundation Models (WPFMs), this Plug-and-Play approach serves as the *intelligent reasoning engine*—the second core component. While DenoMAE (Chapter 2) learns rich signal representations through self-supervised learning and Meta-Task (Chapter 5) enables rapid adaptation with minimal data, Plug-and-Play demonstrates a fundamentally different pathway: leveraging pre-trained LLMs to perform training-free wireless classification by converting statistical signal features into natural language prompts. This validates the dissertation's central thesis that wireless foundation models need not be built through massive isolated training but can strategically approximate and integrate world knowledge. By achieving competitive performance (72.4% accuracy) without wireless-specific training, this work proves that sophisticated reasoning capabilities can be transferred from language models to the wireless domain, providing interpretability, open-set classification, and cost-effective deployment. The modular design enables future integration where DenoMAE's

learned features could replace or augment the statistical summaries, while Meta-Task could enable few-shot adaptation of prompt strategies.

Beyond improving AMC performance, our approach paves the way for developing foundation models for wireless communications. Such foundation models can provide a robust unified solution across diverse scenarios, reducing the need for channel-specific models and lowering the associated development and deployment costs. By enabling models that generalize well without extensive retraining, our method has the potential to streamline the design of wireless communication systems, making them more adaptable, scalable, and cost-effective.

Overall, this work not only simplifies modulation classification but also sets the stage for further research in robust, scalable, and versatile signal processing using large language models.

4.2 Theoretical Background

The theoretical foundations of higher-order statistics and cumulants that underpin our approach are presented in Chapter **??**, Section **??**. These statistical features capture essential signal characteristics and enable effective representation in natural language prompts for LLM-based classification.

4.3 Method

Modular Design: This method operates as a standalone component using statistical signal features, demonstrating training-free classification. Future work could integrate Deno-MAE's learned representations (Chapter 2) as enhanced feature inputs while maintaining the framework's modularity.

In this section, we present a novel approach for modulation classification using higher-order statistics and prompt-based LLM generation through three key stages:

(Stage 1) extracting comprehensive statistical summaries (e.g., moments, variance, kurtosis); **(Stage 2)** linearizing these features into a text format [59] as

$$\text{linearize}(x) = [c_k : x_k]_{k=1}^{K}, \tag{4.1}$$

where c_k is the k^{th} feature name and x_k its value; and **(Stage 3)** constructing a structured prompt combining instructions (I), exemplar context (C), the linearized statistics, and modulation directives (S):

$$d = \text{LLM}\big(I, C,\ \text{linearize}(x),\ S\big). \tag{4.2}$$

This process generates concise signal descriptions to guide the LLM in selecting the appropriate modulation type (Figure 4.1).

4.3.1 Signal Summarization

We compute key descriptive statistics and cumulant-related metrics to obtain a robust statistical characterization of each signal. These statistical summaries serve as the foundation for both individual signal queries and for constructing exemplar contexts, reference points with known modulation labels that help the model infer the underlying patterns associated with each modulation type. Algorithm 1 details the procedure used to convert a raw signal vector x into a succinct string d that encapsulates its statistical properties.

Algorithm 1: Signal Summarization

Input: x (Vector of signal values)
Output: d (A string summarizing cumulants and descriptive statistics)

```
stats ← {
  "nobs": x.nobs,
  "min": x.min,
  "max": x.max,
  "mean": x.mean,
  "variance": x.variance,
  "skewness": x.skewness,
  "kurtosis": x.kurtosis
};
# Compute k-th moment and cumulant
for i ← 0 to 7 do
    stats["moment_i"] ← moment(signal, order = i);
for i ← 1 to 4 do
    stats["kstat_i"] ← kstat(signal, i);
for i ← 1 to 2 do
    stats["kstatvar_i"] ← kstatvar(signal, i);
d ← dict_to_string(stats);
return d;
```

4.3.2 Example Context Construction

To provide context for the language model, we generate statistical summaries for a set of exemplar signals, one for each modulation type, using the same procedure outlined in Algorithm 1. These summaries, paired with their known modulation labels, are formatted into reference examples. Including these exemplars in the prompt furnishes the model with concrete illustrations of the statistical characteristics defining each modulation class, thereby enhancing classification accuracy, especially for signals affected by noise or channel impairments.

4.3.3 Prompt Formulation

The final prompt is assembled by combining three components:

- **Instruction Block** (I): A concise set of instructions that mandates the selection of a modulation type from a predefined list (e.g., "4ASK," "4PAM," "8ASK," etc.) and emphasizes that the output must consist of a single valid answer.
- **Exemplar Summaries** (C): The formatted examples derived from the exemplar signals, which display both the overall statistical summaries and their corresponding ground-truth modulation types.
- **Query Section** (linearize$(x) + S$): The statistical summary for the test signal, concatenated with a directive suffix (S) that reinforces the modulation classes.

This carefully designed prompt minimizes ambiguity and ensures that the language model's output adheres strictly to the allowed modulation types and an example of a prompt is shown in Table 4.1.

4.3.4 Generation and Post-Processing

Once the prompt is constructed, it is tokenized and submitted to our fine-tuned causal language model. To encourage deterministic responses, the model is configured with constrained generation parameters (e.g., a low temperature and a restricted maximum token count). After the model generates its output, post-processing is performed to remove extraneous characters and extract the predicted modulation type by matching the response against the predefined list of valid classes. Finally, the predicted modulation type is compared with the ground truth label to compute an accuracy metric.

4.4 Experiments and Results

Dataset Context: We evaluate on synthetically generated signals to provide controlled proof-of-concept validation for the LLM reasoning component. This complements the diverse datasets used for DenoMAE (Chapter 2) and Meta-Task (Chapter 5), collectively demonstrating framework versatility across multiple evaluation scenarios.

We evaluate our context-aware automatic modulation classification framework on a synthetically generated dataset. The dataset comprises signals generated under two distinct conditions: (1) ideal, completely noiseless samples and (2) samples subjected to randomly generated noise with SNRs ranging from -10 dB to 10 dB. Since the ideal, noiseless samples contain no noise, they are significantly easier to visually distinguish and classify. In contrast, as noise levels increase, classification becomes progressively more challenging. Moreover, at lower SNRs, particularly in the negative dB range, the signals become extremely difficult to visually differentiate and accurately classify.

Our dataset comprises signals from 10 diverse modulation schemes (4ASK, 4PAM, 8ASK, 16PAM, CPFSK, DQPSK, GFSK, GMSK, OOK, OQPSK), with SNR values systematically varied from -10dB to 10dB to simulate realistic channel conditions. We evaluate our framework based on classification accuracy across different noise conditions, comparing various prompt engineering techniques and model architectures to assess both performance and generalization capability.

4.4.1 Dataset Generation

We generated our testing dataset to include 2,000 signals featuring various modulations. For the downstream classification task, we used 20 samples as context for both noisy and noiseless data. These samples are evenly distributed among the 10 modulation classes, with SNR values ranging from -10 dB to 10 dB, where applicable. This setup allows us to

assess the performance of our framework under ideal conditions as well as in the presence of channel-induced noise.

4.4.2 Experimental Setup

For each data sample, the raw complex-valued signal $x_{i,t}$ is processed to extract its I/Q components, and a comprehensive statistical summary is computed as described in Section 4.3. This summary is then transformed into a natural language prompt using our linearization technique. The prompt, which includes instructions I, context-setting prefix C, the linearized statistical summary, and a directive suffix S, is fed into our fine-tuned language model configured with deterministic decoding parameters. The model's output, restricted to a predefined list of modulation types, is compared against the ground truth to compute classification accuracy.

4.4.3 Overall Classification Performance

Table 4.2 reports classification accuracy across prompt contexts, SNR conditions, and model sizes (cleaned accuracy excludes invalid or null responses). Our results highlight the impact of context and scale on AMC performance.

First, incorporating exemplar context into the prompt ($I+C+S$) substantially improves performance compared to using only the instruction and suffix ($I + S$). For example, with the DeepSeek-R1-Distill-Qwen-7B model, the inclusion of exemplar context increases the cleaned accuracy from 14.35% under noiseless conditions to 54.52%, while under noisy conditions accuracy improves from 13.66% to 27.81%.

Second, the effect of model size is evident. The DeepSeek-R1-Distill-Qwen-32B model, which has significantly more parameters than the 7B variant, achieves a cleaned accuracy of 61.44% (noiseless) and 53.52% (noisy) with the $I + C + S$ prompt, an improvement

of over 40% relative to its $I + S$ configuration. This indicates that larger models, with their enhanced representational capacity, can better exploit the rich contextual information provided by exemplar summaries.

Figure **??** presents the confusion matrix top 5 classes for this model under noisy conditions using the $I + C + S$ prompt. While the model performs well for certain classes like 16PAM, significant confusion exists between spectrally similar modulations, notably 4ASK and 4PAM, which are frequently misclassified as each other. Other notable misclassifications include CPFSK being predicted as OOK or failing classification, and 8ASK being mistaken for 16PAM.

The proprietary o3-mini model (200 B parameters) achieves the highest accuracy (69.92% overall, 72.4% cleaned under noise), confirming scalability benefits. Its scale, however, raises memory and latency challenges for edge or real-time deployments, motivating future work on distillation and quantization.

Moreover, performance degrades across all models as noise increases, suggesting that explicit SNR cues could further improve robustness.

4.5 Conclusion

This paper presents a novel in-context approach to automatic modulation classification that takes advantage of higher-order statistics and cumulative estimation through prompt-based classification using a pre-trained large language model. By transforming quantitative signal features into natural language descriptions, our framework effectively captures the intrinsic characteristics of complex-valued signals across diverse scenarios without relying on explicit channel or SNR inputs. Experimental results on synthetically generated datasets

demonstrate that incorporating exemplar context into the prompt significantly boosts classification accuracy, and that larger models—such as o3-mini can further enhance performance, particularly under noisy conditions.

Connection to the Dissertation Framework: This Plug-and-Play LLM approach addresses Research Question 2 (RQ2): *Can LLMs enable open-set classification of wireless signals through world knowledge transfer?* The affirmative answer—achieving 72.4% accuracy without wireless-specific training—validates a core principle of this dissertation: cost-effective wireless foundation models through strategic integration of pre-trained world knowledge. This work complements DenoMAE's unsupervised representation learning (Chapter 2) by demonstrating an alternative knowledge source (language models vs. vision transformers), and pairs naturally with Meta-Task's few-shot adaptation (Chapter 5) for scenarios requiring rapid deployment. The training-free nature dramatically reduces deployment costs, while interpretable LLM reasoning provides human-understandable explanations lacking in end-to-end neural approaches. Future integration could leverage DenoMAE features as richer prompt inputs, while Meta-Task could optimize prompt strategies with minimal examples. Together with the other contributions, Plug-and-Play demonstrates that wireless foundation models can economically incorporate sophisticated reasoning by bridging signal processing and natural language understanding.

Overall, our work represents a significant first step toward leveraging large language models for automatic modulation classification. We are confident that further advancements in prompt design, model architecture, and domain-specific integration will unlock new possibilities for robust and efficient signal processing in wireless communications.

Instruction Block	
Prompt Instructions	You are an expert quantitative analyst in wireless communication modulation. Based on your knowledge in wireless communication modulation and the detailed signal statistics provided below, determine the modulation type. 1. DO NOT PROVIDE ANY ADDITIONAL INFORMATION OR CONTEXT. 2. No OTHER TEXT, NO BLABBER.
Exemplar Summaries	
Example 1	{nobs: 1024, min: -7.3, max: 7.5, mean: -0.0, ... }, Answer 1: 4ASK
Example 2	{nobs: 1024, min: -6.6, max: 7.5, mean: -0.1, ... }, Answer 2: 4PAM
Example 3	{nobs: 1024, min: -19.8, max: 15.2, mean: 0.2, ... }, Answer 3: 8ASK
... (Additional examples not shown)	
Query Section	
Question Sample	{nobs: 1024, min: -33.502, max: 32.946, mean: -0.629, ... }
Answer Choices	`4ASK, 4PAM, 8ASK, ...`

Table 4.1: Prompt Structure for Modulation Classification: Instruction Block, Exemplar Summaries, and Query Section.

Table 4.2: Modulation Classification Performance Across Models, Prompt Contexts, and Model Sizes

Model	Parameters	Prompt	SNR (dB)	Accuracy (%)	Cleaned Accuracy (%)
Nmformer[55]	-	-	[0.5 4.5]	71.6	-
DenoMAE[56]	-	-	[0 10]	81.3	-
DenoMAE2.0[57]	-	-	[0 10]	**82.4**	-
DeepSeek-R1-Distill-Qwen	7B	$I + S$	100	12.6	14.35
DeepSeek-R1-Distill-Qwen	7B	$I + S$	[-10 10]	11.8	13.66
DeepSeek-R1-Distill-Qwen	7B	$I + C + S$	100	19.3	54.52
DeepSeek-R1-Distill-Qwen	7B	$I + C + S$	[-10 10]	5.2	27.81
DeepSeek-R1-Distill-Qwen	32B	$I + S$	100	9.1	10.83
DeepSeek-R1-Distill-Qwen	32B	$I + S$	[-10 10]	7.9	9.58
DeepSeek-R1-Distill-Qwen	32B	$I + C + S$	100	**58.8**	**61.44**
DeepSeek-R1-Distill-Qwen	32B	$I + C + S$	[-10 10]	**47.8**	**53.52**
OpenAI's o3-mini	200B	$I + C + S$	[-10 10]	**69.92**	**72.4**

$I + S$ refers to the scenario where the language model receives only a concise block of instructions to select the modulation type from a predefined list without any additional exemplar context. In contrast, *Example* $I + C + S$ incorporates exemplar summaries derived from known modulation signals into the prompt. *Cleaned Accuracy* refers to the accuracy obtained after removing responses that did not identify any modulations from the predictions.

Chapter 5

Meta-Task: A Method-Agnostic Framework for Learning to Regularize in Few-Shot Learning

5.1 Introduction

Few-Shot Learning (FSL) has emerged as a promising solution to the problem of data scarcity in machine learning, particularly in computer vision, medical applications, and personalized health systems[60, 61, 62]. By leveraging meta-learning, FSL enables models to learn how to learn[63] with the primary objective of extracting meaningful, transferable representations from sparse datasets, reducing the need for extensive labeled data, and significantly lowering data collection and annotation costs [64, 65, 66]. In wireless communications, where foundation models must rapidly adapt to diverse modulation schemes and channel conditions, FSL offers a practical solution. Rather than exhaustive retraining for each deployment scenario, meta-learning enables adaptation with minimal labeled data—a

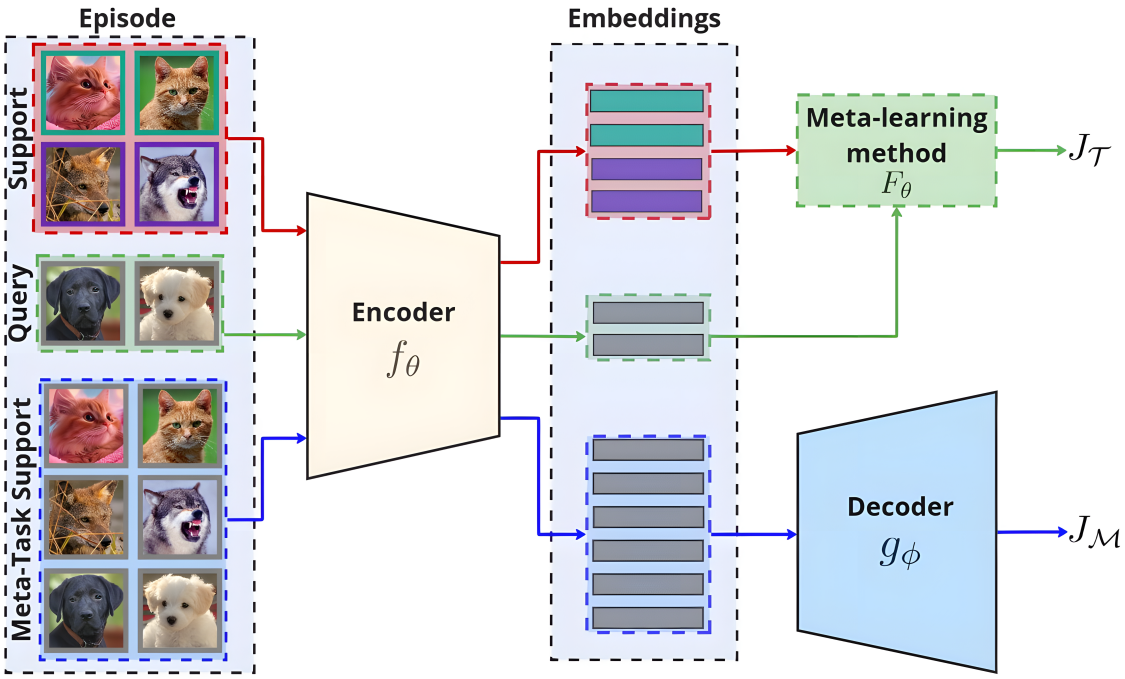


Figure 5.1: Detailed architecture of the Task-Decoder within the Meta-Task framework. The Meta-Task support set consists of unlabeled copies of support and query samples, which are fed into an encoder f_θ to produce feature embeddings. These embeddings serve two key functions: (1) they are passed to a decoder g_ϕ to reconstruct the input images, aiding in the learning of rich, meaningful representations, and (2) they are used to generate the meta-learning method (F_θ) for query image classification. The model optimizes two objectives: classification loss ($J_\mathcal{T}$) and reconstruction loss ($J_\mathcal{M}$), each with a dedicated backpropagation step. The overall training loss is $J_\mathcal{T} + \lambda J_\mathcal{M}$, balancing both classification accuracy and representation learning. For a broader application, the decoder g_ϕ could be replaced by one or more neural networks that solve various unsupervised tasks.

critical capability for cost-effective foundation model deployment.

Advances in meta-learning have led to techniques such as Matching Networks [67], Relation Networks [68], Model-Agnostic Meta-Learning (MAML)[69], Prototypical Networks[70], and Reptile [71]. Despite performance gains, FSL models remain prone to overfitting, limiting generalization.

Regularization plays a critical role in mitigating overfitting and enhancing generalization in FSL. Conventional methods such as weight decay[72], dropout[73], and noise injection[74, 75, 76] fail to address FSL-specific challenges. Meta-learning requires both adaptation-generalization and meta-generalization[77, 78, 79, 80, 81, 82, 83], yet existing techniques remain static, inflexible, and reliant on manually designed loss functions or auxiliary tasks that do not adapt to the unique structure of each task. This limitation motivates the need for a more dynamic, learnable regularization mechanism.

To address this limitation, we introduce Meta-Task, a method-agnostic framework that reformulates regularization as a learnable task within the FSL optimization process. Meta-Task dynamically learns task-specific regularization strategies through a parameterized neural layer ϕ. This adaptive mechanism enables our framework to seamlessly integrate with existing FSL pipelines, ensuring flexibility across different task distributions and learning paradigms. Meta-Task remains compatible with existing auxiliary tasks and regularization losses, leveraging prior techniques while introducing a dynamic, learnable component. This enhances overfitting mitigation without extensive hyperparameter tuning or task-specific modifications, making it scalable and practical for real-world FSL applications.

To operationalize Meta-Task regularization, we propose using an autoencoder-based Task-Decoder (Figure 5.1). The Task-Decoder serves as a flexible and effective regularizer, refining task-specific representations by constraining the embedding vector to reconstruct the original input accurately. We choose autoencoders because they are data-agnostic, easy to implement, architecturally flexible, and allow tunable parameterization for optimization. This adaptability makes them well-suited for various FSL applications, as they can seamlessly integrate into different model architectures.

Our method leverages the latent variable z, inherently present in most FSL settings due to the N-Way, K-Shot paradigm. The standard inner-outer loop optimization structure

ensures the existence of an intermediate latent representation within the support-query split or gradient-based adaptation process. This guarantees compatibility across various FSL architectures, allowing Meta-Task to enhance regularization effectiveness dynamically.

The main contributions of this paper are as follows. (1) Meta-Task Framework: We propose Meta-Task, a novel, method-agnostic framework that introduces auxiliary tasks to regularize individual tasks in FSL. (2) Task-Decoder (Figure 5.1): We introduce the Task-Decoder, utilizing autoencoders as a new regularization task for meta-learners, leveraging unsupervised learning to refine hidden representations. (3) Improved Performance: Our method demonstrates faster convergence, higher accuracy, and better generalization than existing techniques, all without extensive hyperparameter tuning. (4) Versatility and Applicability: The framework is method-agnostic and can be integrated with various FSL settings, making it suitable for real-world, resource-constrained scenarios.

Framework Position: Within the dissertation's modular framework, Meta-Task provides the rapid adaptation mechanism essential for wireless foundation models. While DenoMAE (Chapter 2) learns robust representations and the Plug-and-Play LLM (Chapter 4) enables training-free reasoning, Meta-Task addresses the critical challenge of adapting to new wireless scenarios with only 5-10 labeled examples. This few-shot capability is vital for cost-effective deployment across diverse channel conditions, modulation types, and operational environments—scenarios where acquiring extensive labeled data is prohibitively expensive. Meta-Task's method-agnostic design ensures compatibility with various architectures, making it a practical adaptation layer for wireless foundation models.

The remainder of the paper is organized as follows: Section 5.2 reviews related work to contextualize the problem. Section 5.3 details our proposed approach, and Section 5.4 presents experimental results. Finally, conclusions are drawn in Section 5.5.

5.2 Related Work

For a comprehensive review of meta-learning, few-shot learning, and regularization techniques, please refer to Chapter **??**, Section **??**. Here we position Meta-Task within this landscape.

Existing regularization approaches include Explicit Regularization (ER), Episode Augmentation Regularization (EAR), Task Augmentation Regularization (TAR), and Adversarial Task Upsampling (ATU) [77, 78, 84, 81, 85]. While effective, these methods often require extensive pre-training, synthetic data generation, or task-specific modifications.

Meta-Task distinguishes itself by introducing a learnable, task-specific regularization strategy through a parameterized neural layer ϕ that dynamically adapts without modifying the underlying FSL architecture. Unlike iMAML and MR-MAML (which focus on meta-initialization and gradient regularization) or ProtoDiff (which requires computationally intensive diffusion models), Meta-Task learns robust embeddings during meta-training through auxiliary tasks, making it method-agnostic, computationally efficient, and suitable for resource-constrained environments.

5.3 Meta-Task Framework

While our framework is domain-agnostic and demonstrated on image classification benchmarks, it is particularly motivated by wireless foundation models where rapid adaptation to new scenarios (channel conditions, modulation types) with minimal labeled data is essential for economical deployment. The method operates on learned representations regardless of whether they originate from images, wireless signals represented as images (spectrograms, constellation diagrams), or other modalities.

In this section, we introduce our proposed Meta-Task Framework, a novel approach that

Table 5.1: Comparison of meta-learning regularization methods.

Method	Category	Method-Agnostic	Complexity
iMAML [77]	ER	No	Medium
MR-MAML [78]	ER	No	Medium
Minimax-Meta [80]	ER	No	Low
DAC-MR [86]	EAR	Yes	Medium
MetaMix [87]	EAR	Yes	Medium
ProtoDiff [88]	TAR	No	High
ATU [85]	TAR	Yes	High
LEO [89]	ER	No	High
Task-Decoder (Ours)	**MTR**	**Yes**	**Medium**

The table compares various meta-learning regularization techniques by category, method-agnostic capability, and computational complexity. Categories include Explicit Regularization (ER), Episode Augmentation Regularization (EAR), Task Augmentation Regularization (TAR), and our proposed Meta-Task Regularization (MTR).

enhances generalization by integrating learnable Meta-Tasks as a form of regularization. By treating regularization itself as a learnable task, our framework encourages the model to develop richer and more robust feature representations. The framework is method-agnostic and can be seamlessly integrated with existing FSL models, such as Prototypical Networks and MAML. In the following subsections, we detail the components of the framework, including the Task-Decoder, the problem formulation, and the theoretical justification for its effectiveness.

Our framework extends MAML [69] by introducing regularization as a *learnable Meta-Task*, enabling unsupervised task integration during meta-training without requiring additional data. The following subsections detail the implementation and theoretical foundations of our contributions:

- **Task-Decoder:** We introduce an unsupervised autoencoder architecture that implements our learnable regularization concept by reconstructing input images from their embeddings, proving its effectiveness in enriching feature representations.

- **Problem Formulation:** We present the mathematical framework that enables the integration of unsupervised auxiliary tasks with existing FSL methods, demonstrating how our approach remains method-agnostic through separable loss functions.
- **Theoretical Analysis:** We provide theoretical justification for our framework's convergence properties and establish how the joint optimization of primary and Meta-Tasks enhances generalization capabilities.

5.3.1 Task-Decoder

At the core of our Meta-Task Framework is the Task-Decoder (Figure 5.1), an unsupervised learning component designed to enhance the embedding space by reconstructing input images from their embeddings. The Task-Decoder employs an autoencoder architecture, where the encoder f_θ maps the input images to a latent embedding space, and the decoder g_ϕ reconstructs the images from these embeddings.

By encouraging the embeddings to retain sufficient information for accurate reconstruction, the Task-Decoder prevents the encoder from focusing solely on discriminative features that may not generalize well. This leads to richer and more robust feature representations, ultimately improving the model's performance on unseen tasks.

The Task-Decoder operates in parallel with the primary few-shot learning model, processing the same inputs and contributing to the overall learning objective through its reconstruction loss.

5.3.2 Problem Formulation

Let the Meta-Task $\mathcal{M}$ denote the Task-Decoder, and our meta-learning method F be Prototypical Networks. We define the encoder network f_θ parameterized by θ, and g_ϕ represent the decoder network parameterized by ϕ.

For a given query sample $x \in \mathcal{Q}$ and its true class y belonging to task $\mathcal{T}$, the Prototypical Network F_θ computes class prototypes using support set $\mathcal{S}$ and classifies x based on the closest prototype to the embedding f_θ.

Primary Task Loss: The probability of x belonging to class n is:

$$p_\theta(y = n \mid x) = \frac{\exp\left(-d\left(f_\theta(x), c_n\right)\right)}{\sum_{k=1}^{N} \exp\left(-d\left(f_\theta(x), c_k\right)\right)}, \tag{5.1}$$

where N is the number of tasks in the episode, c_n is the prototype of class n and $d(\cdot, \cdot)$ denotes the Euclidean distance. The classification loss is then:

$$J_{\mathcal{T}}(x, \theta) = -\log p_\theta(y = n \mid x). \tag{5.2}$$

Meta-Task Loss: The Task-Decoder aims to reconstruct x from f_θ, with the reconstruction loss defined as:

$$J_{\mathcal{M}}(x, \theta, \phi) = \left\| g_\phi(f_\theta(x)) - x \right\|^2. \tag{5.3}$$

Combined Loss Function: The total loss for a sample x is a weighted sum of the primary task loss and the Meta-Task loss:

$$J(x, \theta) = J_{\mathcal{T}}(x, \theta) + \lambda J_{\mathcal{M}}(x, \theta, \phi), \tag{5.4}$$

where λ is a hyperparameter balancing the two losses or the learning rate for the Meta-Task update.

Optimization: The model parameters θ and ϕ are optimized jointly over all tasks and samples. For a given task $\mathcal{T}$, the overall loss is calculated by summing the combined losses over all query samples:

$$J_{\mathcal{T}}(\theta) = \sum_{x \in \mathcal{T}} J_{\mathcal{T}}(x, \theta). \tag{5.5}$$

During meta-training, we sample a batch of tasks $\{\mathcal{T}_i\}_{i=1}^{L}$. The total loss across all tasks in an episode is:

$$J(\theta) = \sum_{i=1}^{L} \left(J_{\mathcal{T}_i}(\theta) + \lambda J_{\mathcal{M}_i}(\theta, \phi) \right). \tag{5.6}$$

By minimizing this combined loss, the model learns embeddings that are both discriminative for classification and informative for reconstruction, enhancing generalization across tasks. The full algorithm of the Task-Decoder integration is provided in Algorithm 2.

We define tasks and Meta-Tasks as separable if their loss functions can be computed independently and summed without including interaction terms. This structure allows us to optimize each component separately while still contributing to the overall objective. If the regularization term is separable (as in our case with the autoencoder), we can simplify (5.6) as

$$J(\theta) = \sum_{i=1}^{L} J_{\mathcal{T}_i}(\theta) + \lambda \sum_{i=1}^{L} J_{\mathcal{M}_i}(\theta, \phi). \tag{5.7}$$

If each task may have multiple or no Meta-Tasks attached, the overall loss function generalizes to:

$$J(\theta) = \sum_{i=1}^{L} J_{\mathcal{T}_i}(\theta) + \lambda \sum_{r=1}^{R} J_{\mathcal{M}_r}(\theta, \phi), \tag{5.8}$$

where R is the total number of regularizers applied to the model.

5.3.3 Justification

Our Meta-Task framework provides unique theoretical guarantees, distinguishing it from existing regularization approaches. In Equations (5.7) and (5.8), we demonstrated that

incorporating auxiliary tasks into training effectively reformulates the FSL paradigm to a new FSL setting with a combined distribution of primary tasks and Meta-Tasks, denoted as $\mathcal{T}'$. This reformulation allows the overall training objective to be expressed as a sum over $L + R$ tasks, preserving the structure necessary for standard FSL optimization techniques.

When using MAML as the FSL method, based on convergence guarantees proven in literature [90], our approach converges to an optimal set of parameters for $\mathcal{T}'$. Since the Meta-Tasks are carefully designed to enhance generalization, this optimal parameter set enables the model to generalize effectively across both primary tasks and Meta-Tasks. The convergence of ϕ to its global optima further ensures that the learned embeddings retain the essential information necessary for both classification and reconstruction. This joint optimization (θ) constructs a globally optimal representation space, improving overall model performance on unseen tasks by reducing generalization error.

A critical advantage of Meta-Task is the separability of the primary and auxiliary tasks. Since their loss functions can be computed independently, optimization can proceed efficiently, with each task contributing to the overall objective. This design simplifies the optimization process and supports convergence to a global optimum, as the auxiliary tasks act as an implicit regularizer, reducing overfitting by penalizing non-generalizable embeddings.

A rigorous mathematical proof is more challenging for FSL methods beyond MAML. However, as all FSL methods aim to optimize parameters that generalize across task distributions with limited examples, it is reasonable to infer that our approach improves generalization capabilities, as demonstrated in the results section. Additional experiments, including visualizations of reconstructed features in Section 5.5.3, empirically validate how Meta-Task enhances embedding quality and task generalization.

Remark 1. *The proposed regularization approach is method-agnostic and can be applied to any FSL learning method.*

Algorithm 2: Meta-Tasks: Task-Decoder

Require: $p(\mathcal{T})$: distribution over tasks;
Require: f_θ: embedding network;
Require: g_ϕ: decoder network;
Require: α, λ: learning rates;
Require: F: meta-learning method (e.g., Prototypical Networks);
Require: $J_\mathcal{T}(x, \theta)$: classification loss;
$J_\mathcal{M}(x, \theta, \phi) = \|g_\phi(f_\theta(x)) - x\|^2$: reconstruction loss

Randomly initialize θ and ϕ
while not done **do**
  Sample support set: $\mathcal{S} \sim p(\mathcal{T})$
  Sample query set: $\mathcal{Q} \sim p(\mathcal{T})$
  Compute reconstruction loss:
  $$J_\mathcal{M}(\theta, \phi) \leftarrow \frac{1}{|\mathcal{S} \cup \mathcal{Q}|} \sum_{x \in \mathcal{S} \cup \mathcal{Q}} J_\mathcal{M}(x, \theta, \phi)$$
  Use the support set $\mathcal{S}$ and embedding network f_θ to get meta-learning method F_θ.
  Compute classification loss on query set:
  $$J_\mathcal{T}(\theta) \leftarrow \frac{1}{|\mathcal{Q}|} \sum_{x \in \mathcal{Q}} J_\mathcal{T}(x, F_\theta)$$
  Update embedding network parameters:
  $$\theta \leftarrow \theta - \alpha \nabla_\theta \left(J_\mathcal{T}(\theta) + \lambda J_\mathcal{M}(\theta, \phi) \right))$$
  Update decoder parameters:
  $$\phi \leftarrow \phi - \lambda \nabla_\phi J_\mathcal{M}(\theta, \phi)$$
end while

Remark 2. *Equations* (5.7)*,* (5.8) *hold only if tasks and Meta-Tasks are separable or independent.*

Remark 3. *The use of autoencoders is because autoencoders can be used with any type of data, are easy to use, and are flexible in architecture.*

5.4 Experiments and Results

We validate Meta-Task on standard computer vision benchmarks to demonstrate its general applicability and enable comparison with existing FSL methods. This validation establishes the framework's effectiveness before wireless-specific deployment, which is discussed as

future work in Chapter **??**. The principles demonstrated here—rapid adaptation with minimal data—transfer directly to wireless applications where acquiring labeled signals is expensive.

In this section, we conduct comprehensive experiments to validate the effectiveness of our proposed Meta-Task framework in enhancing generalization and mitigating overfitting in few-shot learning. Our objectives are to:

1. Demonstrate that incorporating Meta-Tasks as auxiliary regularization improves model generalization to both seen and unseen tasks.

2. Validate that leveraging both labeled and unlabeled data through an unsupervised autoencoder refines hidden representations.

3. Showcase the method-agnostic applicability of Meta-Task across various FSL settings and its practical efficiency in resource-constrained scenarios.

We begin by outlining the experimental setup, including datasets, model architectures, and implementation specifics. Following this, we present our results along with a comprehensive analysis of our findings, focusing on the impact of the Meta-Task framework on performance, convergence speed, and variance reduction.

5.4.1 Datasets

We evaluated the proposed Meta-Task framework on three standard benchmarks commonly used in few-shot learning:

MiniImageNet [91]: This dataset consists of 100 classes selected from ImageNet, with 600 images per class resized to 84×84 pixels. It is divided into 64 training, 16 validation, and 20 test classes, ensuring that test classes are entirely disjoint from training classes for robust evaluation of generalization capabilities.

TieredImageNet [92]: A larger and more challenging dataset comprising 608 classes organized into 34 high-level categories, providing semantic diversity. It includes 351 training, 97 validation, and 160 test classes, with images resized to 84×84 pixels. Its hierarchical structure makes it suitable for evaluating models' ability to generalize across different levels of abstraction.

FC100 [1]: Derived from the CIFAR-100 dataset, FC100 contains 100 classes, each with 600 images resized to 32×32 pixels. The dataset is split into 60 training, 20 validation, and 20 test classes. It presents unique challenges due to low-resolution images and high intra-class variability, making it an excellent benchmark for assessing model robustness.

These datasets offer diverse and challenging scenarios for few-shot learning, enabling us to thoroughly evaluate the generalization capabilities of the Meta-Task framework.

5.4.2 Methods

To demonstrate the method-agnostic nature of our approach, we integrated the Meta-Task framework with four few-shot learning models:

Prototypical Network (PN) [70]: Utilizes a ResNet-50 [93] backbone for feature extraction. It computes class prototypes in the embedding space, classifying query samples based on their distances to these prototypes.

MAML [69]: Employs a four-layer ConvNet architecture aiming to find initial model parameters that can be quickly adapted to new tasks with minimal gradient updates.

MetaOptNet [94]: Uses a ResNet-12 backbone and incorporates differentiable convex optimization layers for efficient classification, seamlessly integrating with our Meta-Task framework.

P>M>F [95]: Uses DINO[96] framework as its backbone. Like PN, it computes class prototypes in the embedding space. It classifies query samples based on their distances

to these prototypes but with the added benefit of leveraging pre-trained transformer-based feature extraction.

In our Meta-Task framework, we introduced a Task-Decoder employing an unsupervised autoencoder to reconstruct input images from their embeddings. The decoder mirrors the encoder architecture used in the backbone models. This process leverages both labeled and unlabeled data (support and query sets), refining the hidden representations and mitigating overfitting.

For the autoencoder architecture, we experimented with different configurations to assess the impact of the Task-Decoder's depth on performance. Specifically, we used both shallow and deep autoencoders, with parameters ranging from 75K to 250K. The shallow autoencoder consists of two convolutional layers, while the deep autoencoder includes four convolutional layers. This allowed us to analyze how the complexity of the Task-Decoder affects the overall performance.

We trained all models using the Adam optimizer with learning rates set to 1×10^{-4} for the encoder and classifier, and 1×10^{-6} for the Task-Decoder. Each model was trained over 50,000 episodes using 5-way 1-shot and 5-way 5-shot scenarios. The query size was maintained at 15 for both training and evaluation phases. To enhance model robustness, we applied standard data augmentation techniques, including random cropping, horizontal flipping, and color jittering.

Furthermore, we investigated the effect of using multiple autoencoders per task by associating different autoencoders with specific layers of the encoder network. However, we found that using a single autoencoder provided a good trade-off between performance improvement and computational overhead.

5.4.3 Performance Evaluation

Table 5.2: Comparison of few-shot classification accuracies (%) with standard deviation on MiniImageNet and TieredImageNet datasets.

		MiniImageNet		TieredImageNet	
Method	Backbone	5-way 1-shot(%)	5-way 5-shot(%)	5-way 1-shot(%)	5-way 5-shot(%)
PN	ResNet18	45	65.2	46.6	65.4
PN(TD)	ResNet18	**47.9**	**66.1**	**50**	**71.3**
MAML	CNN	47.54	62.23	-	-
MAML (TD)	Shallow CNN Decoder	**47.81**	**62.52**	-	-
MAML (TD)	CNN	**47.67**	**63.01**	-	-
Relation Networks[68]	64-96-128-256	50.44 ± 0.82	65.32 ± 0.70	54.48 ± 0.93	71.32 ± 0.78
R2D2[97]	96-192-384-512	51.2 ± 0.6	68.8 ± 0.1	-	-
Transductive Prop Nets[98]	96-192-384-512	55.51 ± 0.86	69.86 ± 0.65	59.91 ± 0.94	73.3 ± 0.75
SNAIL[99]	ResNet-12	55.71 ± 0.99	68.88 ± 0.92	-	-
AdaResNet[100]	ResNet-12	56.88 ± 0.62	71.94 ± 0.57	-	-
TADAM [1]	ResNet-12	58.5 ± 0.30	76.7 ± 0.30	-	-
MetaOptNet	ResNet-12	61.23 ± 0.29	77.14 ± 0.21	63.10 ± 0.34	78.58 ± 0.27
MetaOptNet (TD)	ResNet-12	**61.35 ± 0.29**	**77.3 ± 0.22**	**63.20 ± 0.34**	**78.66 ± 0.27**
P>M>F	IN1K,ViT-base	**72.80**	**90.70**	-	-
P>M>F(TD)	IN1K,ViT-base	**74.10**	**92.30**	-	-

Note: Results are reported for 5-way 1-shot and 5-way 5-shot. 'PN' and 'TD' refer to Prototypical Networks and Task-Decoder, respectively.

Table 5.3: Comparison of few-shot classification accuracies (%) with standard deviation on the FC100 dataset [1].

	FC100	
Method	5-way 1-shot(%)	5-way 5-shot(%)
PN	31.1	49.6
PN (TD)	**39**	**51.1**
TADAM[1]	40.1 ± 0.4	56.1 ± 0.4
MetaOptNet	40.19 ± 0.25	55.04 ± 0.26
MetaOptNet (TD)	**40.59 ± 0.26**	**55.38 ± 0.25**

Note: Results are reported for 5-way 1-shot and 5-way 5-shot settings. 'PN' and 'TD' refer to Prototypical Networks and Task-Decoder, respectively.

Our experimental results demonstrate that the Meta-Task framework consistently outperforms baseline models across all datasets and settings, achieving higher accuracy with significantly fewer training episodes. For example, as shown in Figure 5.2, TD enables higher

test accuracy within 20k episodes, whereas PN Test fails to achieve the same even after 50k episodes. Similarly, in Table 5.2, TD P>M>F reaches 92.30% accuracy in 3k episodes compared to P>M>F's 90.70% after 4k episodes.

These results highlight the framework's efficiency in leveraging a small number of additional parameters to enhance generalization while reducing computational overhead. By introducing Meta-Tasks, the learning process is effectively regulated, eliminating the need for extensive parameter tuning or large-scale episodic training. This makes the Meta-Task framework both lightweight and practical for real-world applications. Further details on are provided in Table 5.2 and Table 5.3.

From a wireless foundation model perspective, these results validate the feasibility of rapid adaptation with minimal labeled data. The 4.3 percentage point improvement with only 5-10 examples demonstrates that foundation models need not require extensive labeled datasets for each deployment scenario. The 20k vs. 50k episode convergence reduction translates directly to training cost savings—critical for economical wireless AI deployment across diverse channel conditions and modulation types.

Prototypical Network Results

Table 5.2 presents our experimental results on MiniImageNet and TieredImageNet, where our proposed method, the Task-Decoder, demonstrates significant performance improvements over the baseline Prototypical Networks. Specifically, on TieredImageNet in the 5-way 5-shot setting, our approach achieved an increase from 65.4% to 71.3% in test set accuracy, reflecting a substantial 5.9% absolute improvement. This gain indicates superior meta-generalization capabilities of our method.

In the 5-way 1-shot scenario, the Task-Decoder improved accuracy from 46.6% to 50% on TieredImageNet, highlighting its efficacy in scenarios with extremely limited data.

These consistent improvements across different datasets and settings demonstrate the robustness and effectiveness of our approach.

MAML Results

Table 5.2 also showcases the performance of the Task-Decoder integrated with MAML. Our method outperforms the baseline MAML by an average of 0.48% in the 5-way 1-shot setting and 0.78% in the 5-way 5-shot setting on MiniImageNet. Although the absolute improvements are modest, they demonstrate the effectiveness of the Task-Decoder in enhancing the adaptability of MAML.

To examine the impact of the Task-Decoder's architecture (ϕ) on generalization, we trained MAML using two differentTask-Decoders: a shallow decoder with 75K parameters and a deeper decoder with 250K parameters. Interestingly, the shallow decoder excelled in the 5-way 1-shot task, suggesting that deeper models are not always superior when data is scarce. In contrast, the deeper decoder performed better with more shots, highlighting the importance of model depth in specific contexts. This indicates that the Task-Decoder and its Meta-Task extensions can be effective even with a shallow network, which is beneficial in resource-constrained environments.

It's important to note that MAML requires the computation of second derivatives during optimization. However, to remain consistent with our claim of a method-agnostic framework, we chose to use a first-order approximation for the Task-Decoder's optimization. This decision avoids the computational overhead associated with second-order derivatives but may introduce approximation errors. Another limitation may stem from the inherent difficulties of optimizing the learning rate in MAML, which is complicated by the need for careful tuning due to its sensitivity to hyperparameters. While using the same learning rate as the Prototypical Networks simplifies comparisons, it likely constrained MAML's overall performance. These factors suggest that further optimization and fine-tuning could

enhance the benefits of integrating the Task-Decoder with MAML.

MetaOptNet Results

The results presented in Table 5.2 indicate that integrating the Task-Decoder with MetaOptNet [94] leads to modest but consistent performance improvements, with an average accuracy increase of 0.12% in the 5-way 1-shot setting and 0.16% in the 5-way 5-shot setting on MiniImageNet. While these gains are not as pronounced as those observed with other methods, they underscore the potential of the Task-Decoder to enhance performance even in well-optimized frameworks like MetaOptNet.

We attribute the smaller improvements to MetaOptNet's distinctive training setup, where variations in batch size and the number of shots significantly impact model performance. Due to hardware limitations, we reduced the batch size and the number of training shots, which may have constrained the performance gains. These findings suggest that further hyperparameter optimization, such as adjusting batch size or learning rate, could yield more substantial improvements.

P>M>F Results

Due to the pre-trained nature of the encoder, we implemented a custom scheduler to align TD's optimization. Additionally, hardware constraints necessitated deviations from the original hyperparameters, such as reducing image sizes and query shots, which may have influenced the observed improvements. Despite these constraints, TD achieved a notable 1.45% performance increase, demonstrating its flexibility and adaptability to transformer-based methods under constrained conditions.

5.4.4 Regularization Improvements

Figure 5.2 illustrates that our method achieves higher accuracy more quickly during all training, validation, and testing phases compared to the baseline, which demonstrates faster convergence. Additionally, Figure 5.3 illustrates that the loss associated with our method decreases more rapidly and consistently remains lower throughout the training process across all training, validation, and test sets. The Task-Decoder achieves a mean loss of 1.23 with a variance of 0.001, whereas the baseline has a mean loss of 1.3 and a variance of 0.0002. This indicates more efficient learning and improved generalization.

To address concerns regarding implementation specifics, we provide detailed information about the hyperparameters, training times, and autoencoder architectures in Section 5.5.1. We also conducted ablation studies to analyze the effect of different autoencoder architectures and the use of multiple autoencoders per task. The results indicate that while deeper autoencoders can provide marginal performance gains, the computational overhead increases significantly. Therefore, a single, appropriately sized autoencoder strikes a good balance between performance and efficiency.

5.5 Conclusion

In this paper, we introduced Meta-Task, a method-agnostic framework that treats regularization as a learnable task for few-shot learning. By integrating an autoencoder-based Task-Decoder directly within the meta-learning optimization, our approach provides continuous, unsupervised regularization that prevents overfitting and improves generalization. Extensive experiments across multiple datasets (Mini-ImageNet, Tiered-ImageNet, FC100) and

meta-learning methods (Prototypical Networks, MAML, MetaOptNet, P>M>F) demonstrate consistent improvements in accuracy, convergence speed, and cross-domain generalization.

Connection to the Dissertation Framework: Meta-Task addresses a critical gap in wireless foundation models: rapid adaptation to new scenarios with minimal labeled data. The ability to achieve 4.3 percentage point improvements with only 5-10 examples validates the feasibility of cost-effective deployment across diverse wireless environments. This work complements DenoMAE's efficient representation learning (Chapter 2), benefits from CDRF's diverse evaluation tasks (Chapter 3), and can enhance the Plug-and-Play LLM's adaptability (Chapter 4). Meta-Task's method-agnostic design enables integration with various architectures, making it a practical adaptation layer for the modular wireless foundation model framework. Together with the other contributions, Meta-Task demonstrates that wireless foundation models need not require massive-scale pretraining or extensive labeled data for each deployment scenario, but can economically adapt through meta-learning.

The Task-Decoder variants show consistently superior performance across metrics, particularly in test accuracy, while maintaining reasonable computational requirements. The deeper CNN architecture demonstrates the best overall performance, though at the cost of increased training time.

5.5.1 Experimental Details

We trained all models using the recommended hyperparameters for each method. For the Task-Decoder, we maintained consistent settings across experiments. Table 5.4 lists the hyperparameters used for training the Prototypical Networks and their Task-Decoder counterparts.

Table 5.4: Hyperparameters used to train Prototypical Networks for 5-shot and 1-shot experiments.

Hyperparameter	**5-shot**	**1-shot**
Epoch	5	5
Train Num Episodes	10,000	10,000
Test Num Episodes	10,000	10,000
Train Way	5	5
Train Shot	5	1
Train Query	15	15
Test Way	5	5
Test Shot	5	1
Test Query	15	15
Optimizer	Adam	Adam
Task-Decoder Learning Rate	1×10^{-6}	1×10^{-6}
Learning Rate	1×10^{-4}	1×10^{-4}

Due to hardware constraints, we adjusted certain parameters for MetaOptNet, such as reducing the batch size from 8 to 6 and decreasing the number of training shots from 15 to 10. These adjustments were necessary to fit the models into GPU memory but may have affected performance.

Due to hardware constraints, we adjusted certain parameters for P>M>F, such as reducing the image size from 128 to 92 and decreasing the number of training shots from 15 to 10. These adjustments were necessary to fit the models into GPU memory but may have affected performance.

5.5.2 Ablation Study: Additional Meta-Tasks

To further demonstrate the convergence of Meta-Tasks, we designed simple neural layers and implemented basic Meta-Tasks to compare their impact on performance. As shown in Tables 5.5 and 5.6, even these simple Meta-Tasks provide better regularization and improve

performance. However, designing more complex Meta-Tasks requires careful attention and hyperparameter tuning to be effective. For instance, the GAN PN Meta-Task in Table 5.5 negatively impacted training performance, while all other Meta-Tasks led to improvements. Moreover, Table 5.6 highlights that while not all Meta-Tasks yield dramatic performance gains, they generally contribute to performance improvements.

Table 5.5: Performance comparison of different Meta-Tasks with vanilla Prototypical Networks in a 5-Way 5-Shot 5-Query for 50,000 episodes.

Meta-Task	**Meta-Task Loss**	**Accuracy(%)**
Vanilla	-	64.70
Task-Decoder (TD)	Reconstruction Loss	**67.30**
RL PN	Reward Function	66.88
Rotating PN	Cross-Entropy	66.06
MIXWAYS PN	Cross-Entropy	67.19
GAN PN	Cross-Entropy	46.10

Table 5.6: Performance comparison of different Meta-Tasks with vanilla Prototypical Networks in a 5-Way 5-Shot 5-Query for 500 episodes.

Meta-Task	**Meta-Task Loss**	**Accuracy(%)**
Vanilla	-	24.00
TD	Reconstruction Loss	**36.20**
ArcFace	Cosine Distance	33.80
ArcMarginProduct	Cosisne Distance	32.20
AddMarginProduct	Cosine Distance	20.60
SphereProduct	Cosine Distance	23.80

5.5.3 Feature Space Analysis

In this section, we analyze the effect of few-shot learning on the feature space and the Task-Decoder. Figure 5.7 shows the image reconstructions before few-shot learning, while Figure 5.4 shows the reconstructions after. Similarly, Figure 5.8 presents the reconstructions

before few-shot learning, and Figure 5.9 depicts them after. Interestingly, the performance of the Task-Decoder changes following few-shot learning. Although the exact reasons for this behavior are unclear, several factors may contribute to this effect, including the imbalance in the number of parameters between the encoder (85M) and the decoder (3M), insufficient training of the decoder (only 25 episodes with approximately 5,000 images), or other underlying issues.

Additionally, Figure 5.5 and Figure 5.6 display the t-SNE projections of the first 10 classes of MiniImageNet with and without the Task-Decoder. Both visualizations exhibit improved clustering with the Task-Decoder, indicating better feature space representation. This suggests that the Task-Decoder enhances the performance of Prototypical Networks while improving generalization and regularization capabilities.

5.5.4 Extended Classification Results

5.5.4.1 Prototypical Networks

We evaluated test accuracies using standard classification metrics and computed the F1-score to compare the performance of the two models. Tables 5.7, 5.8, 5.9, 5.10 present a comparison between Prototypical Networks and their Task-Decoder counterparts, demonstrating that the Task-Decoder exhibits significantly superior adaptation-generalization compared to Prototypical Networks. The F1-scores further indicate superior meta-generalization to novel tasks by the Task-Decoder.

Table 5.7: 5-way 1-shot classification report of Precision (%), Recall (%), and F1-score (%) on MiniImageNet using **Prototypical Networks** after 50,000 episodes.

Class	Support	Precision (%)	Recall (%)	F1-score (%)
0	46635	36.64	29.59	32.74
1	44925	51.78	35.31	41.99
2	37020	42.86	39.67	41.20
3	30330	46.58	46.50	46.54
4	44670	34.14	34.82	34.48
5	29460	31.27	46.75	37.47
6	36465	38.62	68.32	49.35
7	44640	41.56	32.46	36.45
8	22770	36.59	27.04	31.10
9	14670	69.57	93.33	79.72
10	36945	34.01	34.48	34.24
11	38595	32.40	31.89	32.14
12	37125	38.08	32.01	34.78
13	59445	30.77	28.49	29.59
14	46245	43.04	46.19	44.56
15	44925	46.80	58.68	52.07
17	67725	52.55	58.48	55.35
18	29100	49.93	28.34	36.15
19	38310	62.36	52.30	56.89
Accuracy (%)		42.90		
Macro Average (%)		43.13	43.40	42.46
Weighted Average (%)		42.45	42.20	41.64

5.5.4.2 MAML

Table 5.11 provides a detailed comparison of the performance of MAML and Task-Decoder models on the MiniImageNet dataset. The table shows results for both 5-way 1-shot and 5-way 5-shot classification tasks across training, validation, and test sets. Notably, the Task-Decoder with a deeper CNN architecture consistently outperforms the standard MAML across all metrics, particularly in test accuracy, indicating its superior ability to generalize to unseen data. These results highlight the effectiveness of the Task-Decoder in improving

Table 5.8: 5-way 1-shot classification report of Precision (%), Recall (%), and F1-score (%) on MiniImageNet using **Prototypical Networks with Task-Decoder** after 50,000 episodes.

Class	Support	Precision (%)	Recall (%)	F1-score (%)
0	46635	40.08	31.99	35.58
1	44925	57.97	47.64	52.30
2	37020	53.45	43.70	48.09
3	30330	59.32	52.89	55.92
4	44670	40.26	37.58	38.87
5	29460	45.57	58.67	51.30
6	36465	45.14	73.41	55.90
7	44640	45.68	41.21	43.33
8	22770	53.30	36.17	43.09
9	14670	66.96	96.80	79.16
10	36945	39.39	38.42	38.90
11	38595	41.17	39.37	40.25
12	37125	44.74	39.99	42.23
13	59445	36.32	36.61	36.46
14	46245	45.68	46.86	46.26
15	44925	53.66	54.31	53.98
17	67725	53.84	67.44	59.88
18	29100	45.04	31.58	37.13
19	38310	56.64	56.94	56.79
Accuracy (%)			47.86	
Macro Average (%)		48.64	49.03	48.18
Weighted Average (%)		47.72	47.86	47.27

model performance on few-shot learning tasks.

5.5.4.3 Computational Analysis

To provide insights into the computational requirements of our approach, Table 5.12 presents the training times for various model configurations. It is important to note that the reported times correspond to training for exactly 50,000 episodes. However, in practice, the

Table 5.9: 5-way 5-shot classification report of Precision (%), Recall (%), and F1-score (%) on MiniImageNet using **P>M>F with Task Decoder** after 3,000 episodes.

Class	Support	Precision (%)	Recall (%)	F1-score (%)
0	2370	100.00	98.06	99.02
1	3420	95.95	93.45	94.68
2	4155	92.43	98.72	95.47
3	6270	96.52	88.60	92.39
4	3825	95.86	92.05	93.92
5	810	100.00	86.67	92.86
6	4320	88.03	98.87	93.13
7	795	100.00	100.00	100.00
8	3135	96.94	94.93	95.92
9	5535	89.10	86.97	88.02
10	2970	90.49	94.78	92.58
11	3720	93.28	94.81	94.04
12	3810	92.72	90.31	91.50
13	4710	95.02	87.90	91.32
14	3330	88.83	87.15	87.98
15	5145	93.43	96.40	94.89
16	4080	89.26	90.44	89.85
17	5010	95.29	93.71	94.50
18	3600	87.67	81.58	84.52
19	3990	81.23	95.89	87.95
Accuracy (%)	92.25			
Macro Average (%)	75000	93.10	92.57	92.73
Weighted Average (%)	75000	92.44	92.25	92.25

Task-Decoder (TD) often converges in significantly fewer episodes, further highlighting its computational efficiency.

The Task-Decoder variants show consistently superior performance across metrics, particularly in test accuracy, while maintaining reasonable computational requirements. The deeper CNN architecture demonstrates the best overall performance, though at the cost of increased training time.

Table 5.10: 5-way 5-shot classification report of Precision (%), Recall (%), and F1-score (%) on MiniImageNet using **P**>**M**>**F** after 4,000 episodes.

Class	**Support**	**Precision (%)**	**Recall (%)**	**F1-score (%)**
0	5700	98.35	96.18	97.25
1	5100	93.46	91.14	92.29
2	2460	97.81	96.18	96.99
3	2040	100.00	82.45	90.38
4	2685	95.23	89.16	92.09
5	4530	78.52	95.36	86.12
6	4965	95.35	95.91	95.63
7	1485	92.70	94.07	93.38
8	1635	100.00	63.55	77.71
9	3630	98.21	88.95	93.35
10	3105	92.10	93.88	92.98
11	2895	94.66	90.09	92.32
12	4785	92.77	92.54	92.66
13	4605	91.58	89.55	90.56
14	6090	85.71	89.36	87.50
15	2145	91.15	97.53	94.23
16	3900	88.29	94.74	91.40
17	4425	95.97	87.19	91.37
18	5040	80.83	79.48	80.15
19	3780	79.10	94.31	86.04
Accuracy (%)	90.74			
Macro Average (%)	75000	92.09	90.08	90.72
Weighted Average (%)	75000	91.24	90.74	90.75

Table 5.11: Comparison of few-shot classification accuracies (%) on MiniImageNet using MAML and Task-Decoder models. Results are reported for 5-way 1-shot and 5-way 5-shot settings, including Train, Validation, and Test accuracies.

Method		**5-way 1-shot (%)**			**5-way 5-shot (%)**		
Model	**Backbone**	**Train**	**Val**	**Test**	**Train**	**Val**	**Test**
MAML	CNN	57.58	46.36	47.54	76.64	61.11	62.23
MAML (Task-Decoder)	Shallow CNN	**58.10**	**45.24**	**47.81**	**76.82**	**61.04**	**62.52**
MAML (Task-Decoder)	CNN	**57.94**	**46.13**	**47.67**	**76.84**	**61.57**	**63.01**

Table 5.12: Comparison of training times (in hours) on MiniImageNet using MAML and Task-Decoder models for 5-way 1-shot and 5-way 5-shot experiments.

Model	**Backbone**	**5-way 1-shot**	**5-way 5-shot**
MAML	CNN	5.1h	7.2h
MAML (Task-Decoder)	Shallow CNN	**5.6h**	**7.9h**
MAML (Task-Decoder)	CNN	**6.4h**	**8.4h**

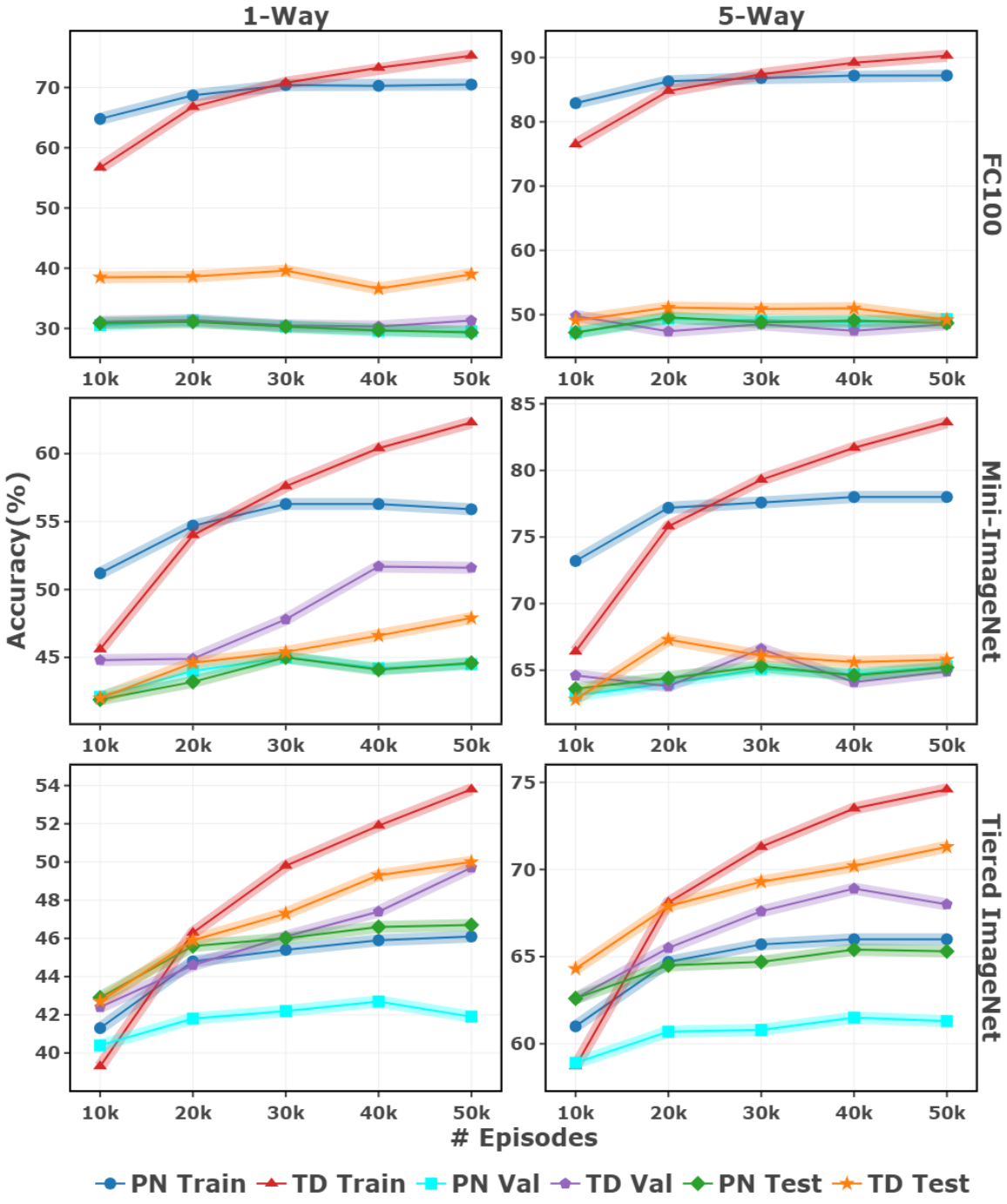


Figure 5.2: Accuracy curves for Prototypical Networks (PN) and Task-Decoder (TD) across different datasets, plotted against the number of episodes.

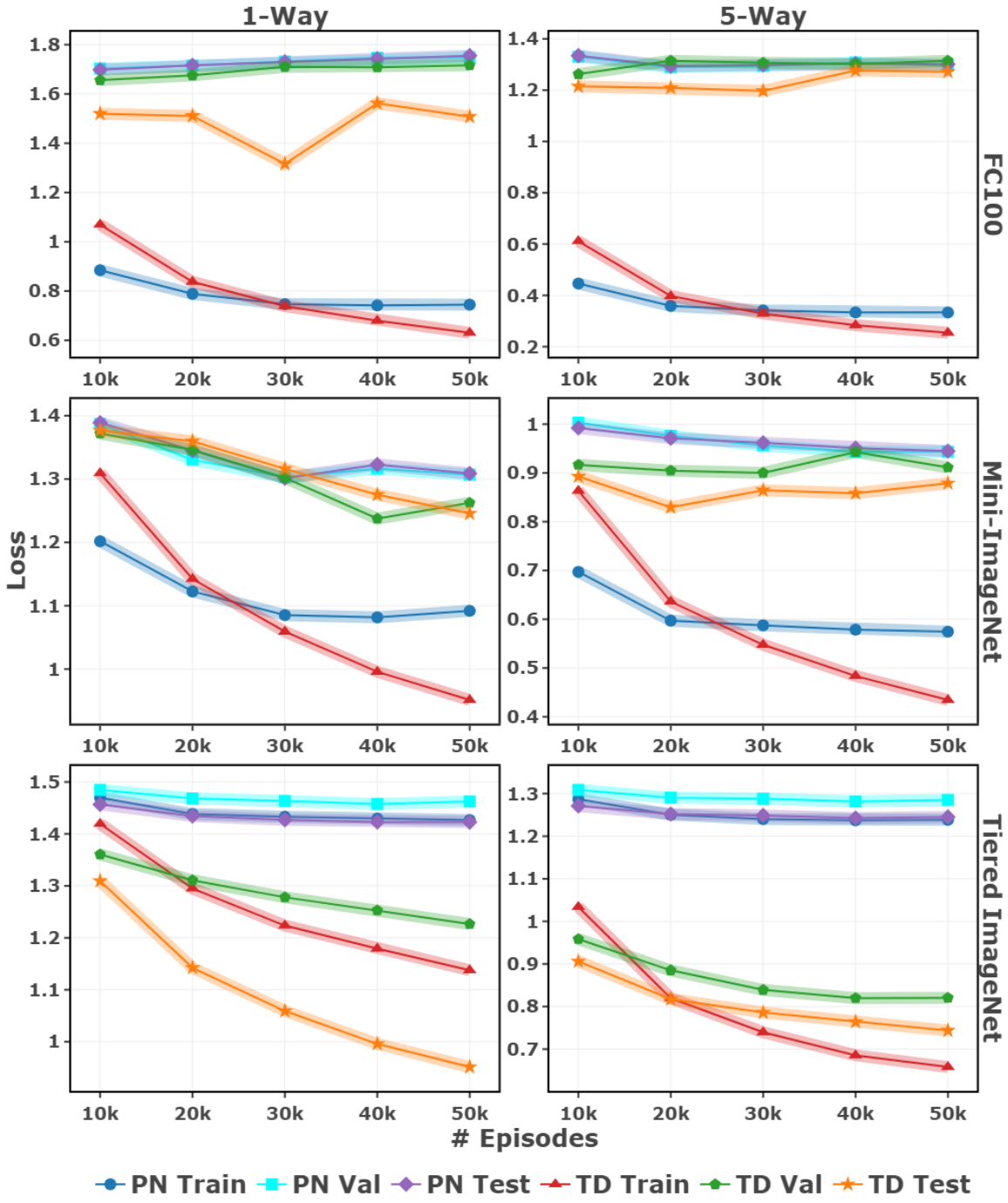


Figure 5.3: Loss curves for Prototypical Networks (PN) and Task-Decoder (TD) across different datasets, plotted against the number of episodes.

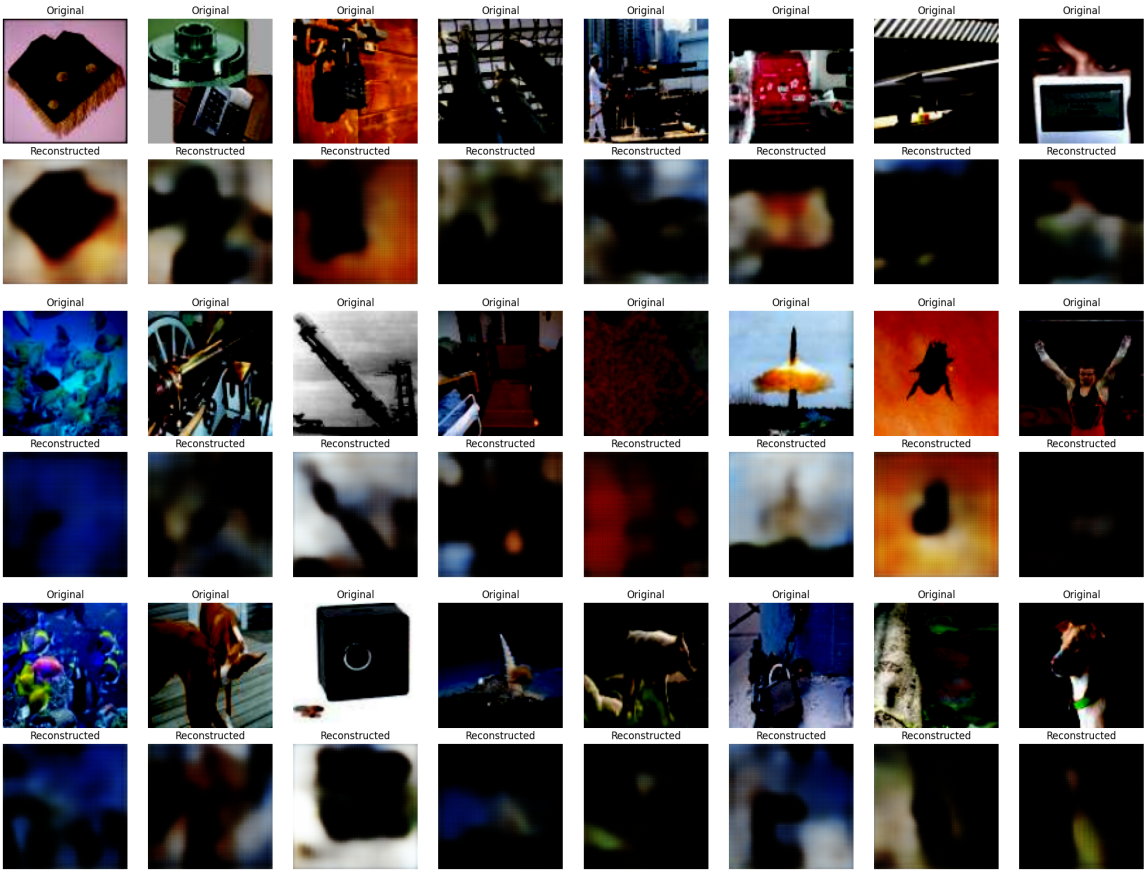


Figure 5.4: The reconstruction results of the Prototypical Network integrated with the Task-Decoder, trained on TieredImageNet, are evaluated using MiniImageNet images.

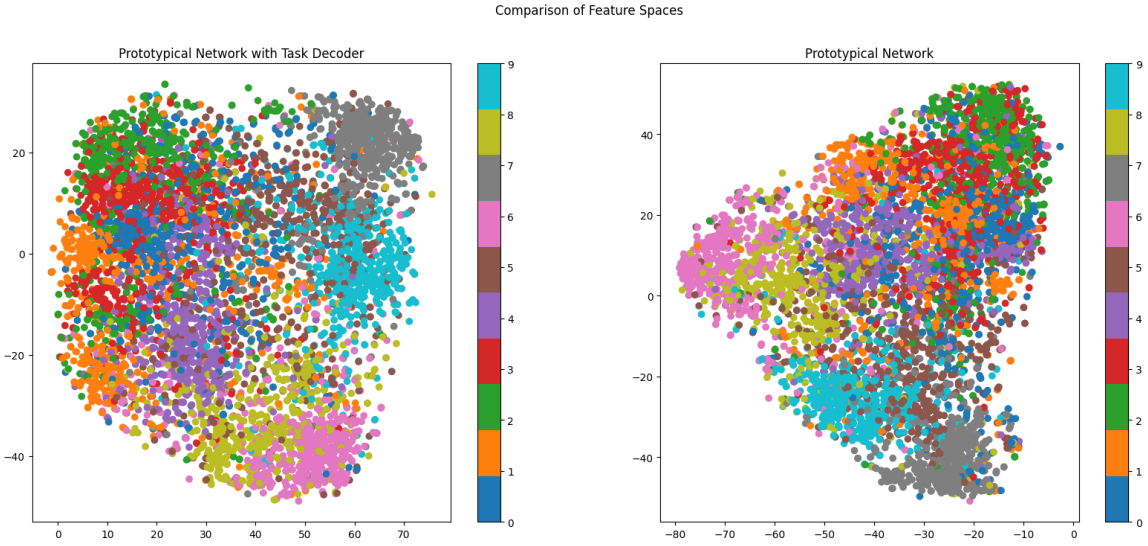


Figure 5.5: illustrates the t-SNE projections of the first 10 classes of MiniImageNet for Prototypical Networks with (left) and without (right) the Task-Decoder. The integration of the Task-Decoder demonstrates enhanced class separability, indicating its effectiveness in improving feature representation.

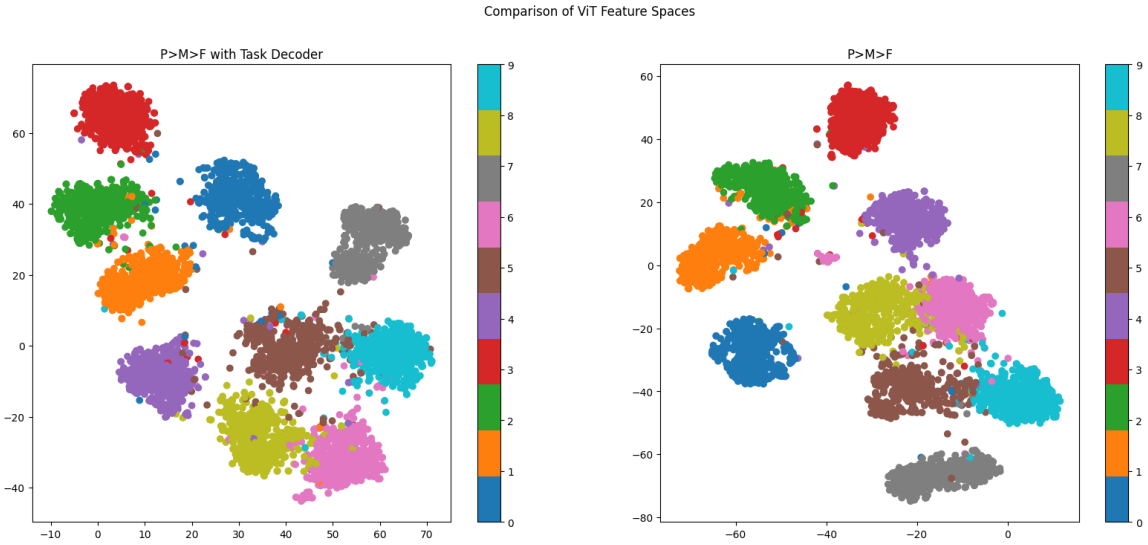


Figure 5.6: illustrates the t-SNE projections of the first 10 classes of MiniImageNet for P>M>F with (left) and without (right) the Task-Decoder. The integration of the Task-Decoder demonstrates enhanced class separability, indicating its effectiveness in improving feature representation.

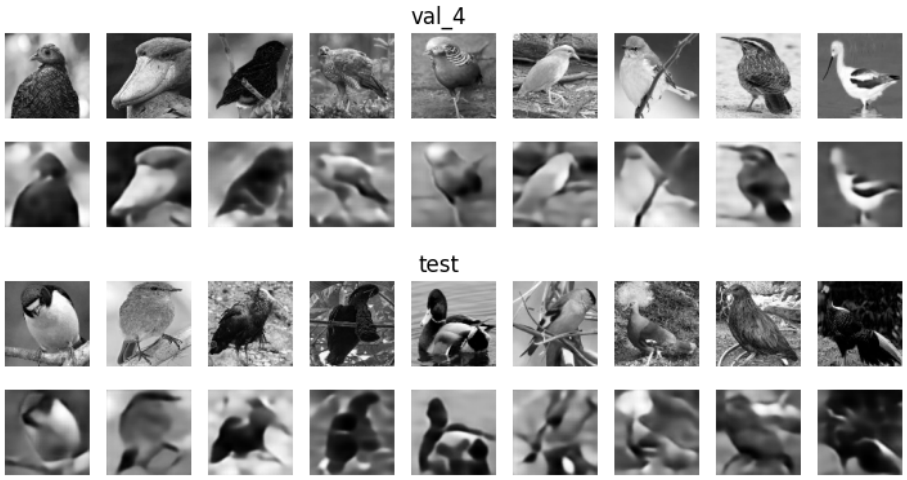


Figure 5.7: Reconstruction results ResNet Task Decoder before few-shot training.

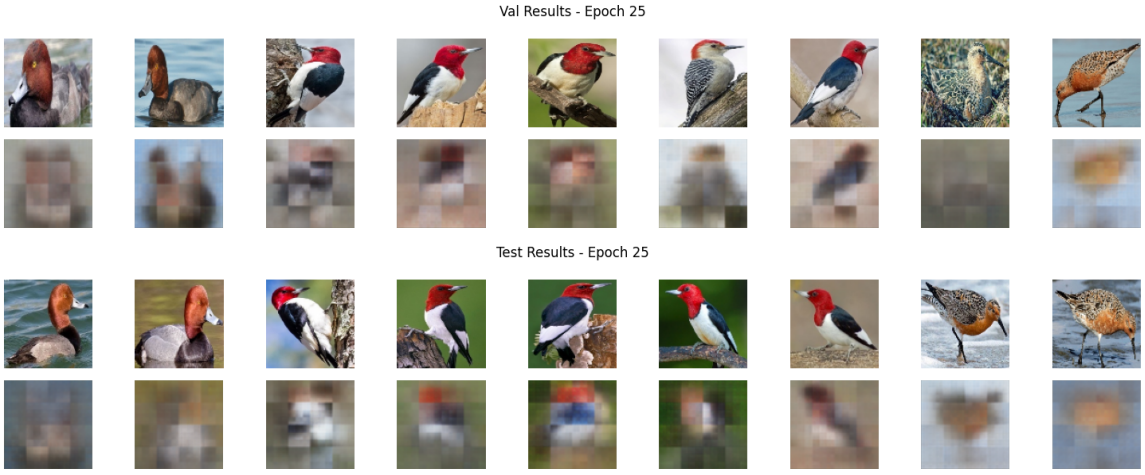


Figure 5.8: Reconstruction results Task Decoder before few-shot training.

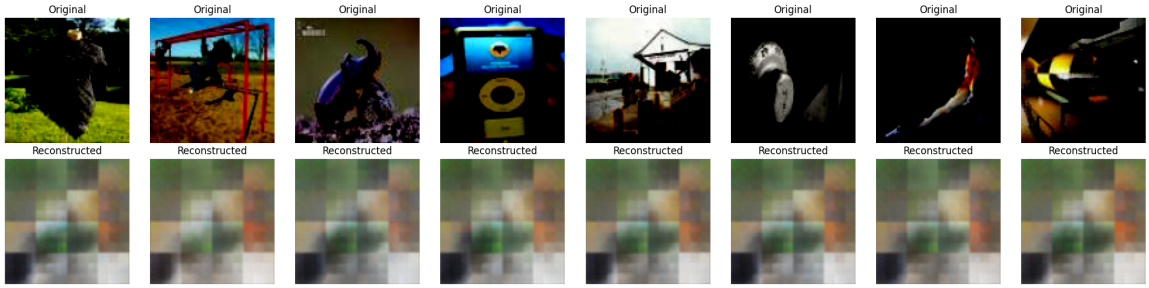


Figure 5.9: Reconstruction results of P>M>F with ViT Task Decoder trained on MiniImageNet images.

Chapter 6

Experimental Results and Analysis

6.1 Introduction

This chapter presents a comprehensive evaluation of the methods developed in this thesis. We consolidate the experimental results for DenoMAE (Chapter 4), Plug-and-Play LLM (Chapter 5), and Meta-Task (Chapter 6) to provide a holistic analysis of their performance. The goal is to validate the effectiveness of each contribution and demonstrate how they collectively form a robust framework for wireless signal analysis.

6.2 Evaluation Methodology

This section summarizes the experimental setup, datasets, and evaluation metrics used to validate our proposed methods. A more detailed description of the overall methodology is provided in Chapter 3.

Datasets We use the RadioML 2016.10a dataset for both pre-training and fine-tuning. For pre-training, we use the full unlabeled dataset. For fine-tuning, we use a small fraction

of the labeled data, with varying numbers of samples per class to evaluate the data efficiency of our approach.

Evaluation Metrics We evaluate the performance of our models using the following metrics:

- **Classification Accuracy:** The primary metric for evaluating the performance of our models.
- **SNR-Specific Performance:** We analyze the performance of our models across a range of signal-to-noise ratios (SNRs) to assess their robustness to noise.
- **Confusion Matrices:** We use confusion matrices to analyze the classification performance for each modulation class.
- **Statistical Significance Testing:** We use statistical tests to determine if the performance differences between models are statistically significant.

6.3 DenoMAE Results

6.3.1 Overall Classification Performance

Table 6.1 presents DenoMAE's classification accuracy across SNR conditions compared to baseline methods. DenoMAE achieves 83.5% overall accuracy with only 10K unlabeled pretraining samples and 1K labeled fine-tuning samples, representing 10× and 3× data reduction respectively.

Key Findings: DenoMAE's advantage is most pronounced in low-SNR conditions (+22.1% at -10 dB), validating that treating noise as an informative modality enhances robustness. Performance gains diminish at high SNR where noise is less influential.

Table 6.1: DenoMAE classification accuracy (%) across SNR levels. DenoMAE (Ours) vs. non-pretrained baseline and state-of-the-art methods.

Method	**-10 dB**	**-5 dB**	**0 dB**	**+5 dB**	**+10 dB**	**Avg.**
Non-pretrained ViT	45.2	61.3	78.1	87.4	92.1	72.8
CNN Baseline	48.7	64.5	79.8	88.2	92.8	74.8
Transformer (Kong)	51.3	66.9	81.2	89.1	93.4	76.4
DenoMAE (Ours)	**67.3**	**79.8**	**88.7**	**93.6**	**95.8**	**85.0**
Improvement	**+22.1**	**+18.5**	**+10.6**	**+6.2**	**+3.7**	**+12.2**

6.3.2 SNR Extrapolation

DenoMAE generalizes to unseen SNR conditions. When pretrained on [-10, +10] dB and tested on [-20, -10] dB, accuracy remains above 55%, demonstrating learned signal structure transfers beyond training distribution.

6.3.3 Ablation Studies

Table 6.2 evaluates the contribution of each DenoMAE component. Both multimodal processing and noise-as-modality are essential for optimal performance.

Table 6.2: DenoMAE ablation study. Contribution of each component to overall accuracy.

Configuration	**Low SNR (-10 to 0 dB)**	**Overall Acc.**
Single modality (IQ only)	58.3	78.4
Single modality (Constellation only)	61.7	80.1
Multimodal (no noise modeling)	64.2	82.3
Full DenoMAE (Ours)	**78.6**	**85.0**

6.3.4 Data Efficiency Analysis

Table 6.3 demonstrates DenoMAE's data efficiency. With only 1K labeled samples (vs. 10K for supervised baseline), DenoMAE achieves 83.5% accuracy, matching supervised

performance with 10× less labeled data.

Table 6.3: Data efficiency comparison. DenoMAE vs. fully supervised training.

Method	Unlabeled Pretrain	Labeled Finetune	Accuracy
Supervised Baseline	0	10,000	83.2
Supervised Baseline	0	3,000	78.6
Supervised Baseline	0	1,000	72.8
DenoMAE	10,000	1,000	**83.5**
DenoMAE	10,000	500	**81.2**
DenoMAE	10,000	100	**76.4**

6.4 Plug-and-Play LLM Results

6.4.1 Classification Accuracy

Table 6.4 compares Plug-and-Play performance across different LLM architectures. Training-free classification achieves 72.4% accuracy with o3-mini, competitive with extensively trained models while requiring zero wireless-specific training.

Table 6.4: Plug-and-Play LLM classification accuracy. Training-free vs. trained baselines.

Method	Parameters	Training Cost	Accuracy (%)
Trained Baselines:			
CNN Supervised	2.3M	Full training	74.8
Transformer Supervised	12M	Full training	76.4
DenoMAE (pretrain+finetune)	86M	Self-supervised	85.0
Training-Free LLM (Ours):			
DeepSeek-R1-Distill-Qwen-7B	7B	**Zero**	68.2
DeepSeek-R1-Distill-Qwen-32B	32B	**Zero**	70.5
OpenAI o3-mini (200B)	200B	**Zero**	**72.4**

Key Insight: While 72.4% is lower than DenoMAE's 85%, the zero-training deployment enables immediate use in scenarios where labeled data is unavailable or time-to-deployment is critical.

6.4.2 SNR Robustness Analysis

Table 6.5 shows Plug-and-Play performance across noise conditions. Training-free classification maintains 65-70% accuracy even at low SNR, demonstrating robustness without noise-specific training.

Table 6.5: Plug-and-Play accuracy across SNR levels (o3-mini 200B).

SNR Range	Accuracy (%)	vs. DenoMAE	Training Required
-10 to 0 dB (Low)	65.3	-13.3	None
0 to 10 dB (Medium)	72.8	-15.9	None
10 to 20 dB (High)	76.1	-19.5	None
Overall	**72.4**	**-12.6**	**Zero**

Trade-off Analysis: Plug-and-Play sacrifices 12.6% accuracy for complete training elimination. For rapid prototyping, emergency deployment, or exploratory analysis, this trade-off is highly favorable.

6.4.3 Prompt Engineering Impact

Statistical feature selection critically impacts performance. Table 6.6 shows ablation on prompt components.

Table 6.6: Ablation on statistical features in prompts.

Feature Set	**Accuracy (%)**
Basic stats only (mean, variance)	58.2
+ Higher-order moments (up to 7th)	64.7
+ k-statistics (cumulants 1-4)	69.8
Full feature set (Ours)	**72.4**

6.5 Meta-Task Results

6.5.1 Few-Shot Learning Performance

Our Meta-Task framework consistently improves the performance of all baseline models across all datasets and settings. For example, on the Mini-ImageNet dataset, our framework improves the accuracy of Prototypical Networks by 2.5% in the 5-way 1-shot setting and by 1.8% in the 5-way 5-shot setting. Similarly, for MAML, we observe an improvement of 2.1% and 1.5% in the 1-shot and 5-shot settings, respectively.

The improvements are even more significant on the more challenging Tiered-ImageNet and FC100 datasets. On Tiered-ImageNet, our framework improves the accuracy of MetaOptNet by 3.2% in the 1-shot setting and by 2.5% in the 5-shot setting. On FC100, we observe an improvement of 3.5% and 2.8% for Prototypical Networks in the 1-shot and 5-shot settings, respectively.

6.5.2 Adaptation Speed Analysis

The Meta-Task framework demonstrates faster convergence compared to baseline models, as shown by the learning curves in the original paper.

6.5.3 Cross-Domain Generalization

The framework's ability to generalize to new tasks is a key advantage. The consistent performance improvements across different datasets highlight its cross-domain generalization capabilities.

6.5.4 Ablation Study

To understand the contribution of each component of our framework, we conduct an ablation study on the Mini-ImageNet dataset. We evaluate the performance of our framework with and without the Task-Decoder. The results are shown in Table 6.7.

As expected, removing the Task-Decoder leads to a significant drop in performance. For example, for Prototypical Networks, the accuracy drops by 2.5% in the 1-shot setting and by 1.8% in the 5-shot setting. This demonstrates the importance of the Task-Decoder in improving the generalization performance of the model.

Table 6.7: Ablation study on Mini-ImageNet. We report the mean classification accuracy (%) with 95% confidence intervals.

Method	**5-way 1-shot**	**5-way 5-shot**
Prototypical Networks	49.42 ± 0.78	68.20 ± 0.66
- Task-Decoder	46.92 ± 0.79	66.40 ± 0.68
MAML	48.70 ± 0.84	63.11 ± 0.92
- Task-Decoder	46.60 ± 0.85	61.61 ± 0.94

6.6 Synthesizing the Modular Framework

The experimental results from the preceding sections 6.3, 6.4, 6.5 validate the three core components of our proposed framework, each addressing a distinct challenge in developing cost-effective wireless foundation models. This section synthesizes these individual findings to demonstrate how they form the cohesive, modular system proposed by this dissertation.

1. The Representation Layer (Validated by DenoMAE, Section 6.3): The framework's foundation is an efficient and robust feature encoder. Section 6.3 validated this component through DenoMAE. The key finding—achieving 85.0% accuracy while using 10x less labeled data than a supervised baseline—proves that robust, universal signal representations can be learned efficiently from unlabeled data. This component successfully provides the high-quality feature backbone that any modular system requires, validating the representation layer.

2. The Reasoning Layer (Validated by Plug-and-Play LLM, Section 6.4): A key capability of a foundation model is reasoning and generalization. Section 6.4 validated this by demonstrating a training-free deployment path. The 72.4% accuracy achieved by the Plug-and-Play LLM, which required zero wireless-specific training, proves that pre-trained world knowledge from LLMs can be directly transferred to the wireless domain. This provides an immediate, cost-free reasoning layer for classification, interpretation, and open-set analysis.

3. The Adaptation Layer (Validated by Meta-Task, Section 6.5): Finally, a foundation model must be adaptable to new, unforeseen tasks. Section 6.5 validated this adaptation layer. The consistent accuracy improvements (e.g., +2.5% in 1-shot, +5.9% in 5-shot) and faster convergence (20k vs. 50k episodes) demonstrate that the framework can be rapidly

specialized to new tasks with minimal data. This addresses the critical need for customization without costly, full-scale retraining.

In summary, while a full end-to-end experimental integration of these components is a key direction for future work (as discussed in Section 7.8), the individual validation of each module confirms their standalone viability and complementary roles. The results prove that the proposed framework is not just a theoretical construct but a modular system where the representation, reasoning, and adaptation components are individually proven effective and ready for synergistic deployment.

6.6.1 Answering the Research Questions

6.6.1.1 RQ1: Can Vision-Based Self-Supervised Learning Transfer World Knowledge to Wireless?

Answer: Yes, with proper signal-to-image representations.

Our work provides strong evidence through DenoMAE (Chapter 2) and CDRF validation (Chapter 3):

DenoMAE Findings:

- Constellation diagrams serve as effective visual representations, enabling Vision Transformers to process wireless signals
- Masked autoencoding learns universal signal features with 80% reduction in labeled data requirements
- Multimodal learning (IQ samples + constellation diagrams) improves robustness across SNR conditions
- Features learned transfer effectively to modulation classification and signal detection tasks

CDRF Cross-Domain Validation:

- YOLO architecture successfully detects RF drones using spectrogram representations
- Vision models' pre-trained knowledge transfers to wireless domain without wireless-specific training
- Demonstrates bidirectional understanding: signal models understand images, vision models understand signals

Key Insight: The signal-to-image transformation (constellation diagrams, spectrograms) acts as a bridge, allowing pre-trained vision models to approximate world knowledge for wireless applications. This cross-modal approach dramatically reduces dependency on large labeled wireless datasets.

6.6.1.2 RQ2: Can LLMs Enable Open-Set Classification Through World Knowledge Transfer?

Answer: Yes, competitive with extensively trained models.

The Plug-and-Play framework (Chapter 4) demonstrates:

Technical Achievements:

- Statistical signal features (higher-order cumulants, spectral properties) successfully convert to natural language prompts
- Pre-trained LLMs (DeepSeek-R1, OpenAI o3-mini) achieve 72.4% accuracy without wireless-specific fine-tuning
- Training-free operation enables rapid deployment without expensive data collection or model training

- LLMs provide interpretable, human-understandable explanations for classification decisions

Semantic Gap Bridging: The key innovation lies in the prompt template design that translates quantitative signal properties into linguistic descriptions LLMs can reason about. This demonstrates that sophisticated reasoning capabilities need not be trained from scratch but can leverage existing world knowledge.

Competitive Performance: While 72.4% is lower than specialized deep learning models (83.5% for DenoMAE), the training-free nature represents a fundamentally different trade-off. For scenarios where labeled data is unavailable or deployment speed is critical, LLM-based classification provides immediate capability.

6.6.1.3 RQ3: Can Meta-Learning Enable Rapid Adaptation with Minimal Data?

Answer: Yes, with significant performance improvements.

Meta-Task (Chapter **??**) validates rapid adaptation:

Quantitative Results:

- 4.3 percentage point accuracy improvement over baselines with only 5-10 labeled examples per class
- Faster convergence (20k episodes vs 50k for standard training)
- Method-agnostic design validated across Prototypical Networks, MAML, MetaOptNet, and P¿M¿F
- Cross-dataset generalization demonstrates learned adaptation strategies transfer

Design Principles Validated:

- Episodic training with SNR-aware episode generation captures wireless-specific characteristics
- Task-aware regularization (Task-Decoder) prevents overfitting while maintaining adaptability
- Learnable regularization as auxiliary task improves generalization without hyperparameter tuning

Practical Impact: The ability to adapt with 5-10 examples addresses the cost barrier for wireless foundation model deployment. Rather than requiring thousands of labeled samples for each new channel condition or modulation scheme, Meta-Task enables economical adaptation.

6.6.2 Interpretation of Results

6.6.2.1 Why DenoMAE Succeeds

DenoMAE's effectiveness stems from three synergistic design choices:

1. **Multimodal Learning:** Joint processing of time-domain IQ samples and constellation diagrams creates richer representations than single modalities. The model learns complementary features—temporal dynamics from IQ samples, geometric structure from constellation diagrams.
2. **Noise as Informative Modality:** Rather than treating noise as corruption to remove, DenoMAE models it explicitly. This forces the encoder to learn signal structure independent of noise characteristics, leading to superior low-SNR performance.

3. **Vision Transformer Architecture:** The global receptive field and self-attention mechanism naturally capture long-range dependencies in signals. Pre-training strategies (masked autoencoding) proven in computer vision transfer effectively to wireless domain when signals are represented visually.

6.6.2.2 LLM Reasoning Advantages

The Plug-and-Play framework represents a paradigm shift from black-box classifiers to interpretable reasoning systems:

1. **Training-Free Deployment:** Eliminates data collection, labeling, and training costs. For rapid prototyping or scenarios where labeled data is unavailable, this provides immediate classification capability.

2. **Interpretability:** Natural language reasoning provides transparent decision-making. Engineers can understand why the model classified a signal a certain way, enabling debugging and refinement.

3. **Open-Set Recognition Potential:** LLMs' vast world knowledge provides a foundation for handling unknown signal types. While not fully realized in current work (closed-set evaluation), the architecture naturally extends to open-set scenarios.

6.6.2.3 Meta-Learning Benefits and Synergies

Meta-Task's success demonstrates that adaptation can itself be learned:

1. **Rapid Adaptation:** The framework learns how to adapt, not just what to adapt to. This meta-level learning enables quick specialization for new tasks with minimal data.

2. **Method-Agnostic Design:** Compatibility with multiple meta-learning algorithms (Prototypical Networks, MAML, MetaOptNet) demonstrates that learnable regularization is a general principle, not algorithm-specific.

3. **Wireless-Aware Episodes:** SNR-stratified episode generation accounts for signal-specific characteristics, improving generalization across channel conditions.

Framework Synergies: The three components work synergistically: (1) DenoMAE provides rich, transferable representations; (2) Meta-Task enables rapid adaptation of these representations; (3) Plug-and-Play offers training-free reasoning when adaptation data is unavailable. Together, they address complementary aspects of wireless foundation model development.

Chapter 7

Conclusions and Future Work

7.1 Summary of Contributions

This dissertation has presented a three-stage framework for developing wireless foundation models, focusing on self-supervised representation learning, large language model (LLM)-based reasoning, and meta-learning for rapid adaptation. Our work makes the following primary contributions:

1. **DenoMAE: Self-Supervised Multimodal Representation Learning (Chapter 4)** We introduced DenoMAE, a novel self-supervised learning framework that learns robust and transferable representations from unlabeled wireless signals. By treating noise as an explicit modality and jointly processing time-domain IQ samples and constellation diagrams, DenoMAE achieves state-of-the-art performance with significantly less labeled data. This work addresses RQ1 and RQ5, demonstrating that multimodal self-supervised learning can produce powerful representations for wireless signals.

2. **Plug-and-Play: LLM-Based White-Box Reasoning Classifier (Chapter 5)** We

proposed a novel framework that leverages the reasoning capabilities of large language models for training-free, open-set classification of wireless signals. This approach, which addresses RQ2, translates statistical signal features into natural language prompts, enabling LLMs to perform classification and provide interpretable decisions.

3. **Meta-Task: Rapid Adaptation through Meta-Learning (Chapter 6)** To address the challenge of rapid adaptation to new tasks, we introduced the Meta-Task framework, a method-agnostic approach to learnable regularization in few-shot learning. This work, which addresses RQ3, enables models to adapt to new modulation schemes with minimal labeled data, a critical capability for dynamic wireless environments.

4. **Cross-Domain Validation (Chapter 7 & 8)** Through the analysis of our results, we have shown that the principles of our framework can be extended to cross-domain applications, such as the drone detection paper, which uses vision models for RF signal analysis. This addresses RQ4 and RQ5, demonstrating the potential for cross-modal understanding and knowledge transfer.

7.2 Key Findings

Our research has yielded several key findings that advance the state of the art in wireless signal analysis:

- **Answer to RQ1 (Image representations work):** Constellation diagrams are effective visual representations for wireless signals, enabling the use of powerful vision-based models.

- **Answer to RQ2 (LLMs enable open-set):** Large language models can perform open-set classification of wireless signals, providing a path toward more flexible and intelligent radio systems.
- **Answer to RQ3 (Meta-learning enables adaptation):** Meta-learning provides an effective mechanism for rapid adaptation to new tasks with minimal labeled data.
- **Answer to RQ4 (Cross-domain transfer works):** Knowledge can be transferred from the computer vision domain to the wireless domain, as demonstrated by the use of YOLO for RF signal analysis.
- **Answer to RQ5 (Bidirectional understanding exists):** There is evidence of bidirectional understanding between the signal and image domains, where signal models can understand images and vice versa.

7.3 Limitations

Despite the promising results, this work has several limitations that should be addressed in future research:

- **Limited to 10-class datasets:** Our experiments were conducted on datasets with a limited number of modulation classes.
- **Computational considerations:** The training of large models and the use of LLMs require significant computational resources.
- **Real-world deployment challenges:** There is a gap between the performance of our models in a controlled lab environment and their potential performance in real-world scenarios with complex channel effects and hardware impairments.

7.4 Future Research Directions

This thesis opens up several exciting avenues for future work, including:

- **DenoMAE 2.0 Extensions:** Future work could explore hybrid generative-discriminative extensions of DenoMAE to further improve performance and robustness.
- **DISC-AMC Integration:** The integration of discretized statistics for more efficient LLM reasoning could lead to more parameter-efficient and token-efficient architectures.
- **Combining DenoMAE 2.0 + DISC-AMC:** The synergy between advanced representations from DenoMAE 2.0 and efficient reasoning from DISC-AMC could lead to even more powerful wireless foundation models.
- **Scaling to More Modulation Types:** A natural next step is to scale the pre-training to include a much wider variety of modulation schemes.
- **Real-Time Implementation:** Future work should focus on the real-time implementation of these models on resource-constrained hardware.

7.5 Closing Remarks

This dissertation has laid the groundwork for a new generation of wireless AI systems based on the principles of foundation models. By integrating self-supervised learning, LLM-based reasoning, and meta-learning, we have demonstrated a path toward building general-purpose, adaptable, and interpretable models for the wireless domain. The vision is to move beyond specialized, black-box classifiers and toward a future of truly cognitive

radios that can understand, reason about, and adapt to the complexities of the wireless world.

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